NAVAL AIR TRAINING COMMAND

NAS CORPUS CHRISTI, TEXAS **CNATRA P-1288 (Rev. 03-25)**

FLIGHT TRAINING INSTRUCTION

TACTICAL FORMATION T-45 STRIKE

2025

DEPARTMENT OF THE NAVY
CHIEF OF NAVAL AIR TRAINING
250 LEXINGTON BLVD SUITE 179
CORPUS CHRISTI TX 78419-5041

CNATRA P-1288
N715
26 Mar 2025

CNATRA P-1288 (Rev. 08-24)

Subj: FLIGHT TRAINING INSTRUCTION, TACTICAL FORMATION T-45 STRIKE

1. CNATRA P-1288 (Rev. 08-25) PAT, "Flight Training Instruction, Tactical Formation T-45 Strike" is issued for information, standardization of instruction, and guidance to all flight instructors and students in the Naval Air Training Command.

2. This publication will be used as a guide for completion of the T-45 STRIKE curricula for all Student Naval Aviators (SNA).

3. Recommendations for changes should be submitted via the electronic Training Change Request (TCR) form on the Chief of Naval Air Training (CNATRA) Web site.

4. CNATRA P-1288 (Rev 08-24) PAT is hereby cancelled and superseded.

S. V. LAMOURE
By direction

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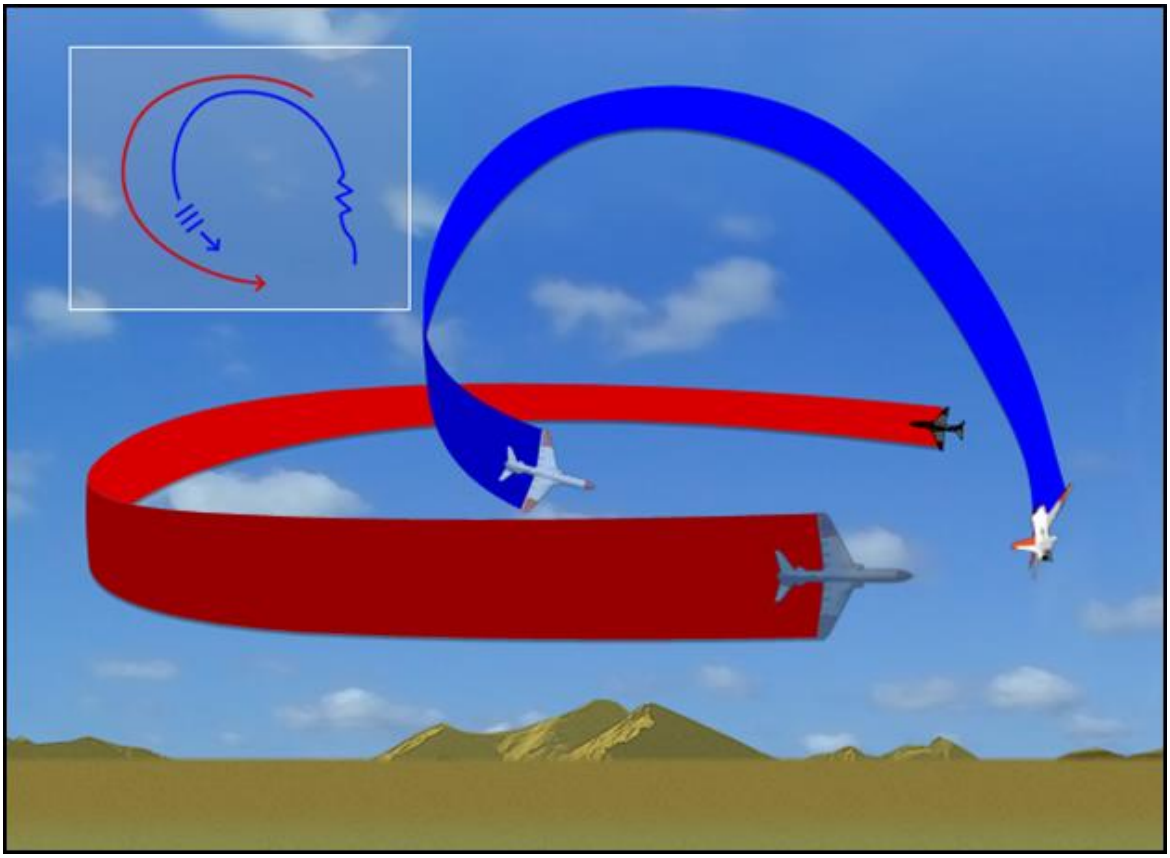
FLIGHT TRAINING INSTRUCTION

FOR

TACTICAL FORMATION

T-45



LIST OF EFFECTIVE PAGES

Dates of issue for original and changed pages are:
Original. . . 10 Oct 08 (this will be the date issued)
Revision…1…10 Aug 09
Revision…2…09 Nov 12
Change…1…03 Aug 17
Revision…3…11 May 20
Revision…4…23 Jan 23
Revision…5…30 Mar 23
Revision…6…26 Sep 24
Revision…7 26 Mar 25

TOTAL NUMBER OF PAGES IN THIS PUBLICATION IS 58 CONSISTING OF THE FOLLOWING:

Page No.	**Change No.**	**Page No.**	**Change No**
COVER	0		
LETTER	0		
iii –viii	0		
1-1 – 1-14	0		
2-1	0		
2-2 (blank)	0		
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3-6 (blank)	0		
4-1 – 4-4	0		
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7-1 – 7-7	0		
7-8 (blank)	0		

INTERIM CHANGE SUMMARY

The following Changes have been previously incorporated in this manual:

CHANGE NUMBER	REMARKS/PURPOSE

The following interim Changes have been incorporated in this Change/Revision:

INTERIM CHANGE NUMBER	REMARKS/PURPOSE	ENTERED BY	DATE

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CHAPTER ONE
TACTICAL FORMATION

100. INTRODUCTION

Combat Spread is a mutually supportive formation in which both fighters work together to maximize their lethality and survivability in the visual arena. This stage will provide the basic procedures for maneuvering a section in Combat Spread.

101. CONCEPTS AND DEFINITIONS

Environmentals

All flight members shall assess and record the environmentals (sun azimuth and elevation, cloud decks in operating area, and winds at altitude in operating area) and assess how those conditions affected the flight conduct.

Combat Spread

Combat Spread enables a section to maintain a high degree of maneuverability while also preserving a high degree of visual mutual support. Wing will fly on Lead's 90° bearing line (abeam or 3/9 line) at a distance of .8 to 1.0 nm, and 1,000 ft above or below Lead's altitude at 330 KIAS. Lead will typically brief 15,000 ft for Lead, and 16,000 ft for Wing during the syllabus TAC FORM flights. The two Fighters are spaced far enough abeam to still be maneuverable but allowing aircrew the ability to devote more time to cockpit tasks and visual lookout. Realize that although Wing may fly a stepped-up position in training here, the option to fly stepped down may be used due to environmentals and various other reasons.

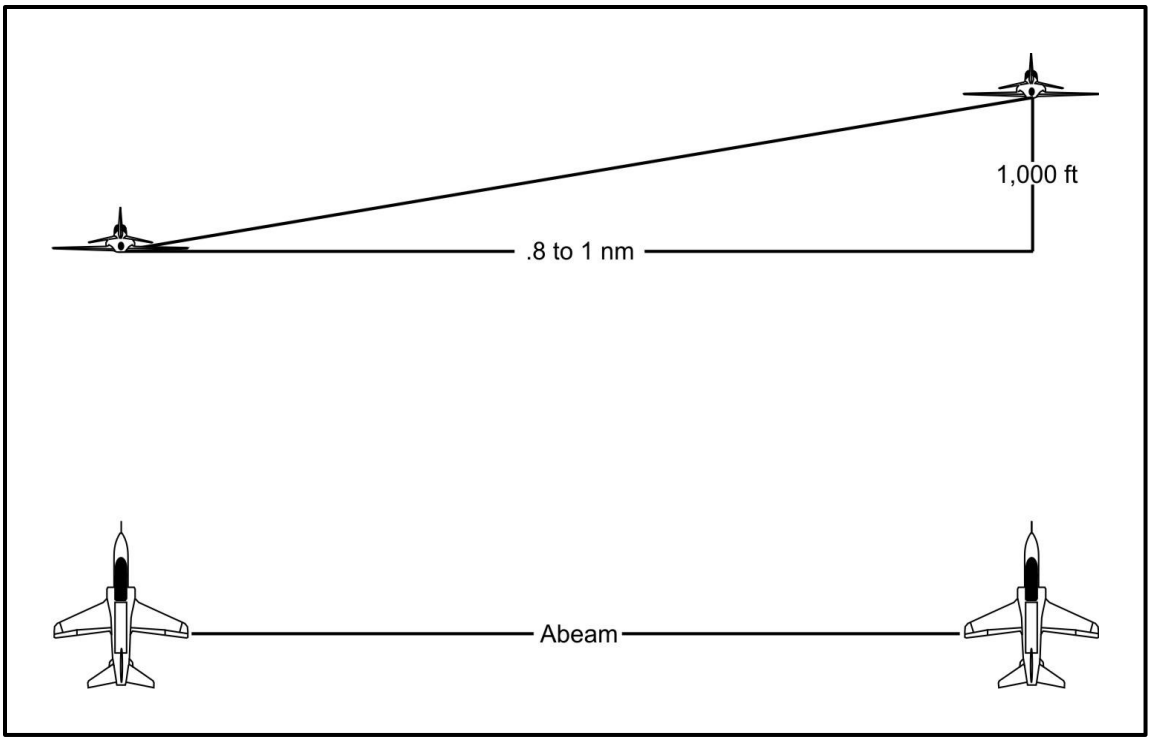

Figure 1-1 Combat Spread

Tactical Wing Position

Tac Wing position is used during the Tac Admin portion of the flight to allow Lead flexibility to maneuver the section as required to a briefed location, execute a join, or to facilitate pushing Wing to a briefed tactical formation. Increased distance between Wing and Lead (greater than Admin Cruise) allows Lead to maneuver more dynamically but also allows Wing increased mission cross-check time to complete cockpit tasks. Tac Wing for the T-45 is defined as Lead's 45° to 60° bearing line, 0.2-0.6 nm, and no lower than 500 ft below Lead's altitude with flexibility to move from side to side. Attempt to avoid large throttle movements and capitalize on the fluidity of the position to utilize geometry to flow to the outside and inside of the turn as required to maintain position. Upon completion of a maneuver in Tac Wing, Wing should move to a position that is visible to Lead. Tac Wing position and maneuvering will be utilized in follow on stages of training and in the fleet.

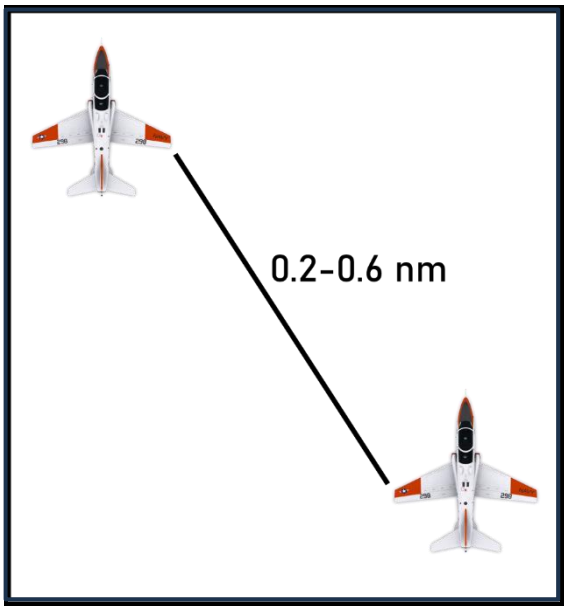

Figure 1-2 Tac Wing Position

"Visual/Tally" and "Blind/No Joy"

'Visual' and 'Blind' refer to whether or not a fighter can see a friendly aircraft. 'Tally' and 'No Joy' mean the same thing respectively but are applied to adversary aircraft.

Engaging Turns

An Engaging Turn is an energy-sustaining turn performed at 300 or 330 KIAS, Maximum Rated Thrust (MRT), G to maintain airspeed. It allows the section to be maneuvered quickly without an energy penalty.

Hard Turns

A Hard Turn is a compromise between turn performance and energy depletion. Hard turns are flown at 17 units AOA (nibble of buffet), 300 KIAS, MRT, and are used to quickly re-orient the section and/or regain visual mutual support following an engagement.

Break Turns

A Break Turn is a maximum performance turn executed at 19-21 units AOA (roughly the maximum lift a T-45 can generate), but not to exceed 7.33 Gs (T-45 G-limit above 5,000 ft). A break turn gives the best instantaneous turn performance, but at a large energy penalty. A break turn may be used offensively to gain a shot opportunity, or defensively to defeat an incoming missile or a Bandit's firing solution. It is important to note that max-performing the aircraft does NOT mean to use erratic or jerky aircraft control inputs. Encountering pitch buck in the T-45 is common; if this occurs, ease off the G, and then smoothly reapply G as required until within the rate airspeed band.

Significant Airspeeds

The T-45's Best Sustained Turn Rate Airspeed Band is 240-330 KIAS. At the lower end of the rate band the aircraft yields a small advantage in turn rate (0.5° higher at 240 KIAS, compared to 330 KIAS). The T-45's ability to trade airspeed for angles (energy excursion) is however reduced below 300 KIAS; at slower airspeeds the aircraft becomes more difficult to handle as it is more prone to stall/pitch buck. ***Initially, targeting the upper portion of the rate band (300-330 KIAS), until an energy excursion is necessary, is prudent in most cases.***

Lift Vector

The Lift Vector is an imaginary line that extends upward (relative to pilot's body), and perpendicular to the plane of the wings, regardless of pitch or roll angle. In the T-45, the lift vector is visualized by looking straight up between the canopy MDC cords. If inverted, the lift vector would be pointing toward the ground. In a pure vertical climb, the lift vector would be on the horizon.

Angle-Off-Tail (AOT)

Angle-Off-Tail is a descriptive position relative to an aircraft's longitudinal axis, with 0° being directly behind the aircraft's tail, and 180° being directly ahead of the aircraft's nose. Intermediate angles between 0° and 180° are not specific to just the top, bottom, or sides of an aircraft; any AOT can be described at any one point around an aircraft that lies on a cone of that angle.

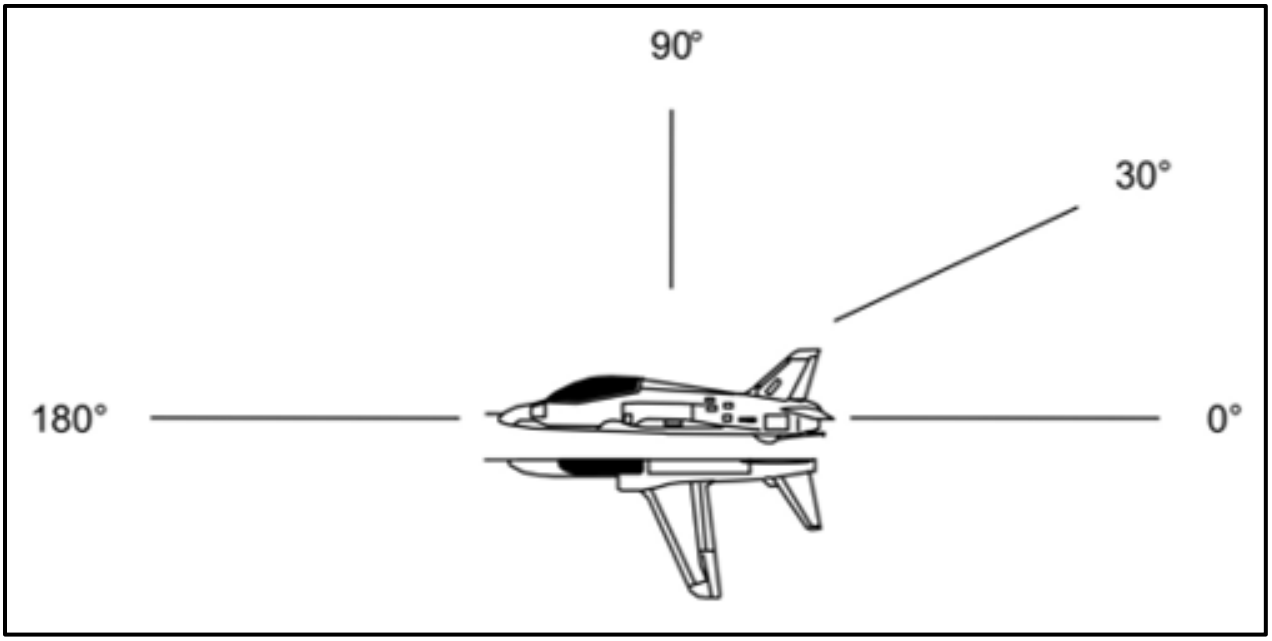

Figure 1-3 Angle-Off-Tail (AOT)

Aspect

Aspect quantifies (in general terms) where an aircraft is positioned relative to another aircraft, such as the Bandit, and is generally defined by three separate ranges of AOT with respect to the other aircraft. Low aspect arises when looking, more or less, at another aircraft's tail (0-45 deg AOT); medium aspect as we approach the other aircraft's planform or wing line (45-90 deg AOT); and high aspect is forward of the other aircraft's planform or wing line (90-180 deg AOT). Aspect change describes how an object appears to pivot.

Line of Sight (LOS)

Line of sight is just what it sounds like. It's the line from a pilot's eyeball to an object. 'LOS rate' describes how an object appears to move across the canopy.

Closure

Closure is the rate at which separation between two aircraft is decreasing. A negative closure would describe a separation that is increasing or opening.

Pursuit Geometry

The concept of pursuit geometry between attacker and defender is basic to every tactical maneuver. The type of pursuit can be determined by the attacker's nose position when in the same plane of motion as the defender, and lift vector placement when not in the same plane of motion. There are three basic types of pursuit:

1. Lead pursuit: used to decrease nose-to-tail separation, or solve for a gun solution, often at the cost of generating high AOT and closure.

2. Pure pursuit: used primarily to gain an IR missile lock. Pure pursuit is achieved by putting the nose (velocity vector) on the bandit. While not as rapid as lead pursuit, pure pursuit can be used to close nose-to-tail distance if the pursuing fighter is at least co-airspeed with the bandit.

3. Lag pursuit: used to maintain or increase nose-to-tail separation. Lag pursuit allows the Fighter to maintain energy while keeping AOT and closure under control.

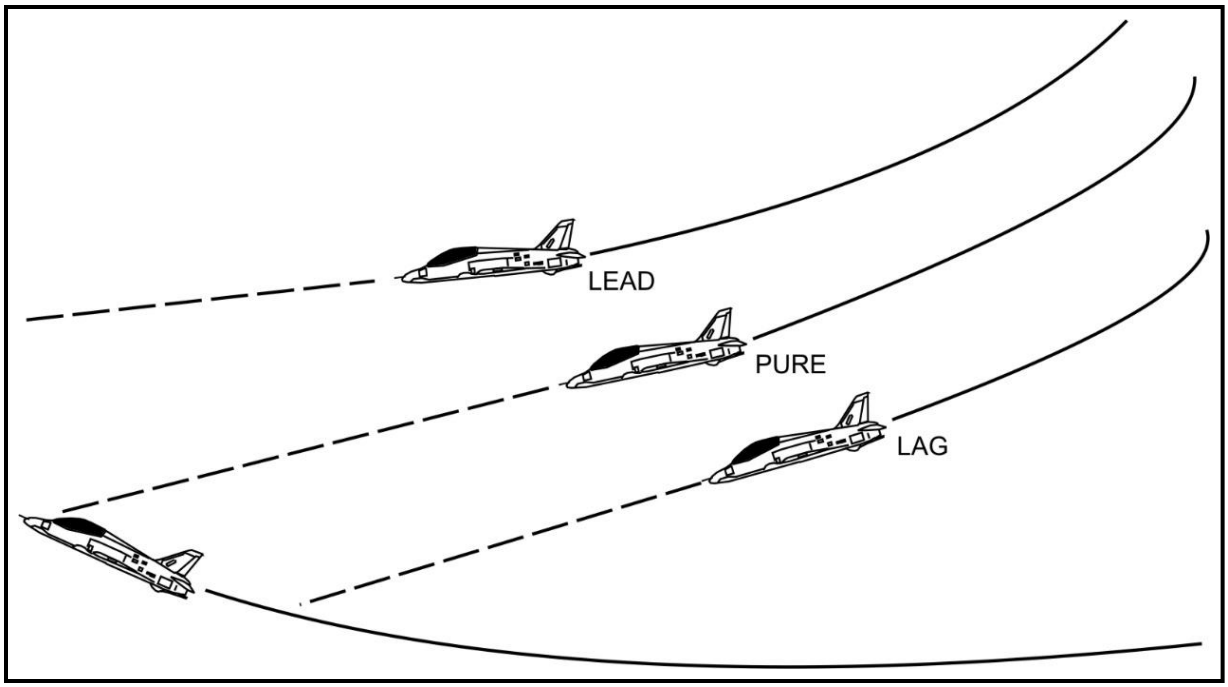

Figure 1-4 Pursuit Curves In-Plane

We can expand the concept of pursuit geometry from the two-dimensional discussion above, to three dimensions. For example, an attacker that is flying in a different plane of motion can orient the ***lift vector*** ahead of, on, or behind a defender to achieve many of the same goals as would be achieved with nose position if flying in the same plane of motion.

Turn Circle

A Turn Circle is an imaginary two-dimensional circle that an aircraft is scribing through the sky, defined by the aircraft's performance parameters at any instance in time. In a horizontal turn, the turn circle is, more or less, just a circle. In the vertical plane, such as an aircraft in a constant-G loop, gravity will shrink an aircraft's turn radius at the top and widen its turn radius at the bottom.

Bubble

The Bubble is a three-dimensional representation of an aircraft's turn circle in a max performance turn. The Bubble should not be confused with the 500 ft bubble that Wing needs to maintain from Lead at all times during dynamic maneuvering. The average Bubble of the T-45 is represented by a circle with a radius of 3,000 ft.

Post

The Post is the center point of a turn circle. This is the same post encountered during a TACAN rendezvous but is not restricted to turn circles in the horizontal plane. The Post does not define pursuit curves, but is used in the discussion of the Bubble, Attack Window Entry (AWE), etc.

Control Zone (CZ)

The Control Zone is a truncated cone extending from 2000 ft to 4000 ft behind the defending aircraft, and along its flight path. The volume is defined along this length expanding from 20° wide at the front, to 40° wide at the back. The practical application of the Control Zone is such that should an attacking aircraft arrive within a defending aircraft's Control Zone, ***with range, angles and closure under control***, the defending aircraft has little chance of forcing the attacking aircraft out of the Control Zone.

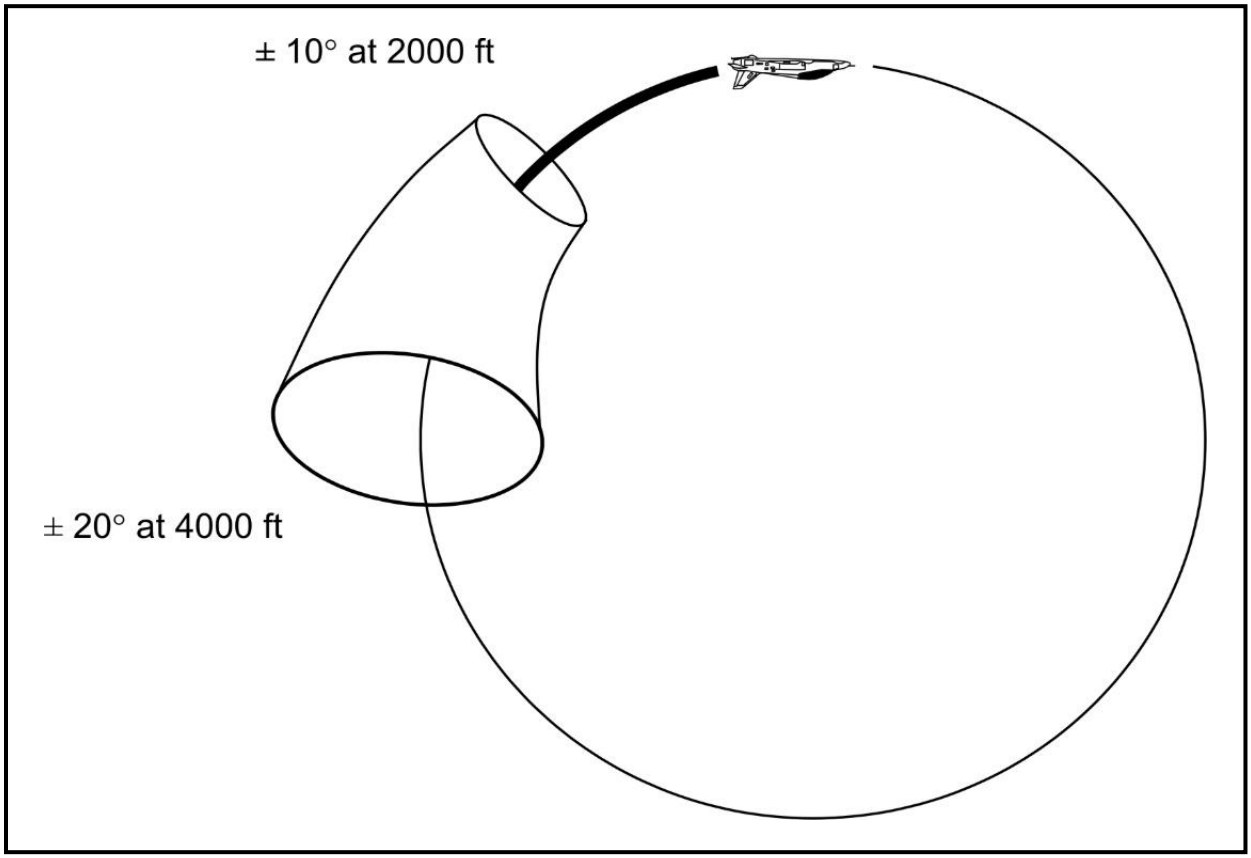

Figure 1-5 Control Zone

Attack Window

The Attack Window is the point in the sky where, if an attacking aircraft executes a max performance turn, it will arrive in the defensive aircraft's Control Zone (with the aforementioned range, angles, and closure under control). ***The most useful cue for the Attack Window Entry is a sudden decrease in the Bandit's aspect change, accompanied by a sudden increase in the Bandit's LOS rate.*** In practical terms, the Bandit's aircraft appears to stop pivoting and begins to rapidly move aft across the canopy.

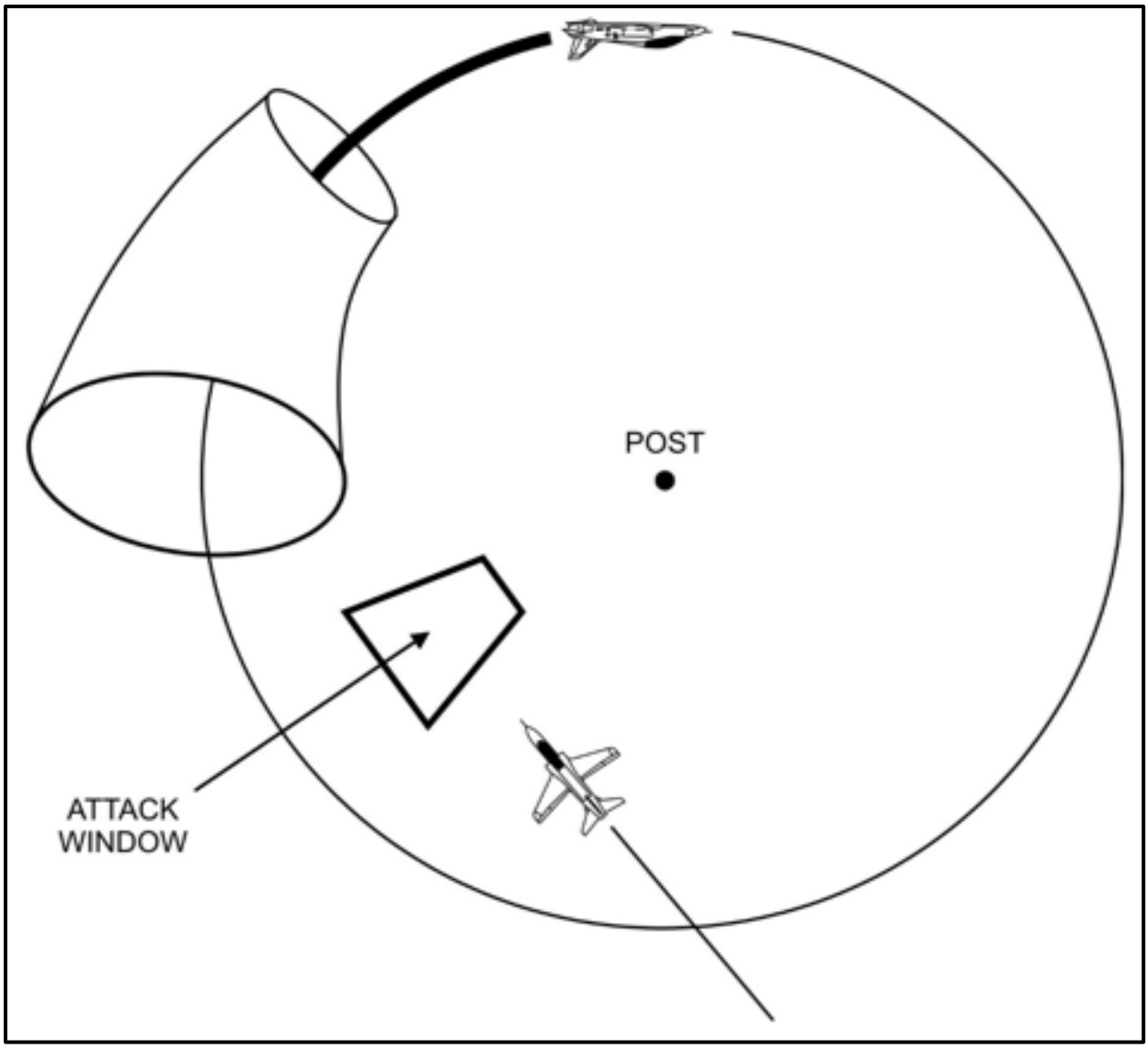

Figure 1-6 Attack Window

Misaligned Turn Circles (MATC)

MATCs are created due to the fact that each aircraft's bubble is displaced over a different geographic point in space. Given the same exact turn performance, the geometry of MATCs will enable an attacker's nose to eventually come to bear on the defender through pure geometry without having to perform an energy excursion.

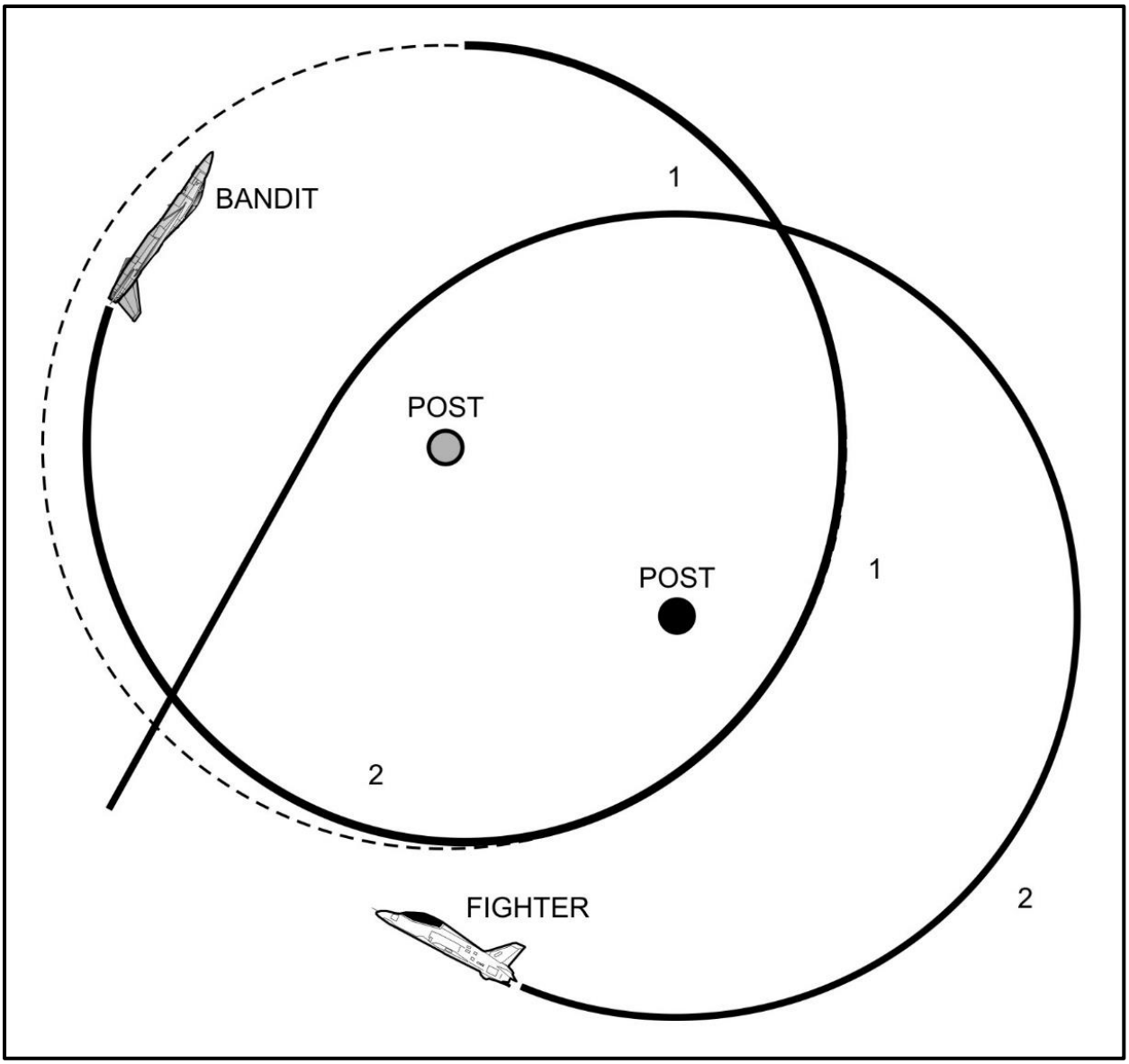

Figure 1-7 Misaligned Turn Circles

Low Yo-Yo

The Low Yo-Yo is a nose-low (and possibly out-of-plane) maneuver used to decrease nose-to-tail separation.

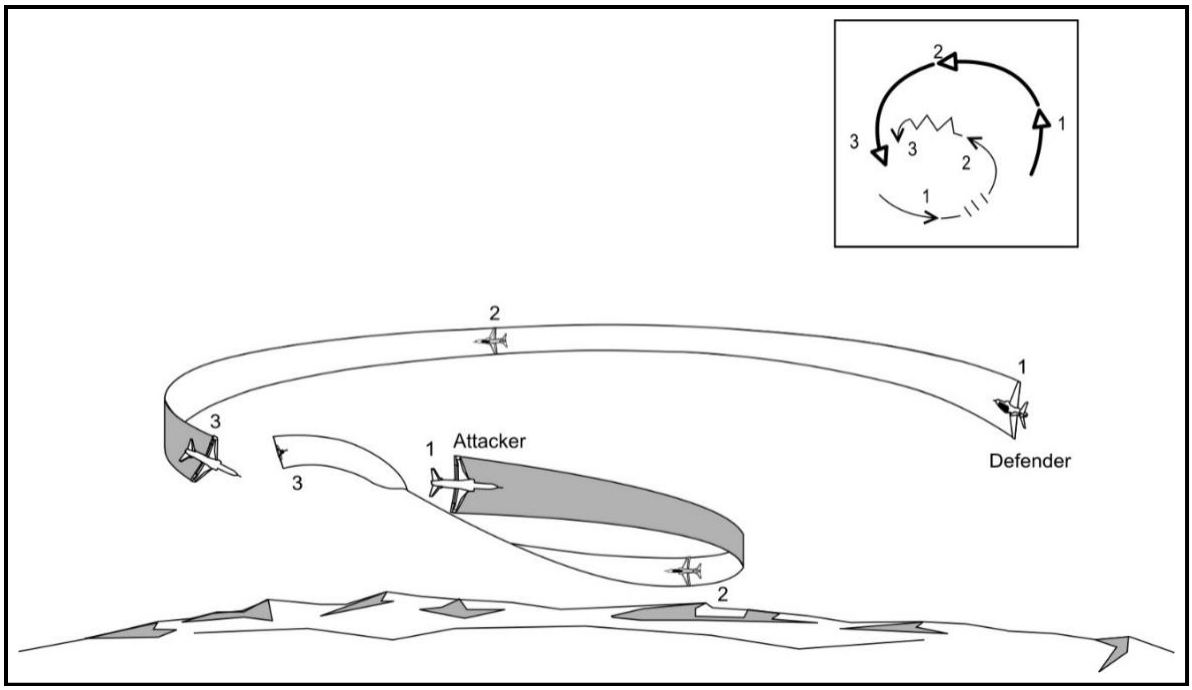

Figure 1-8 Low Yo-Yo

High Yo-Yo

The High Yo-Yo is a nose-high (and again, possibly out-of-plane) maneuver to slow or stop closure and/or increase nose-to-tail separation.

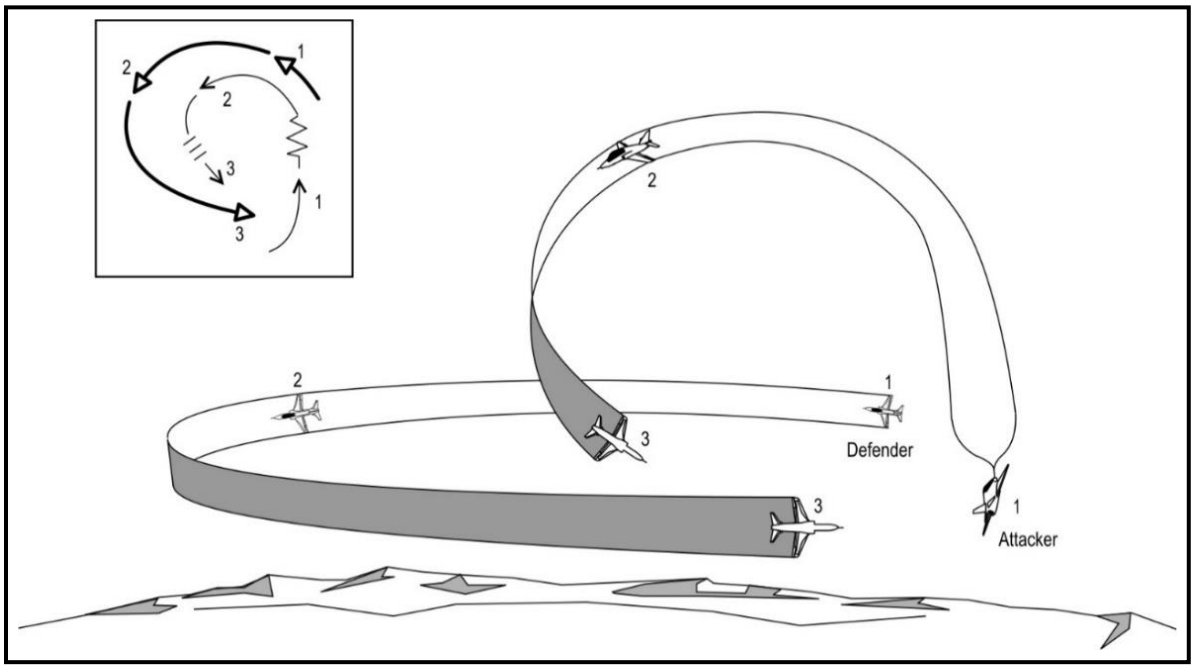

Figure 1-9 High Yo-Yo

Three Overshoots:

1. Flight path overshoot – flying through the adversary's flight path. Ensure the flight path overshoot occurs within or aft of the confines of the CZ. This will deny a reversal opportunity.

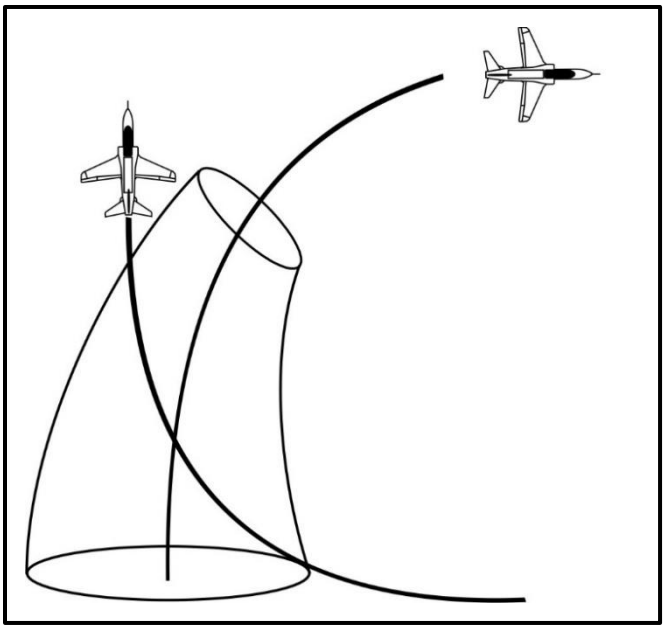

Figure 1-10 Flight Path Overshoot

2. In-close overshoot – a flight path overshoot that occurs inside the near portion of the defender's CZ, within 2,000 ft aft of the defender.

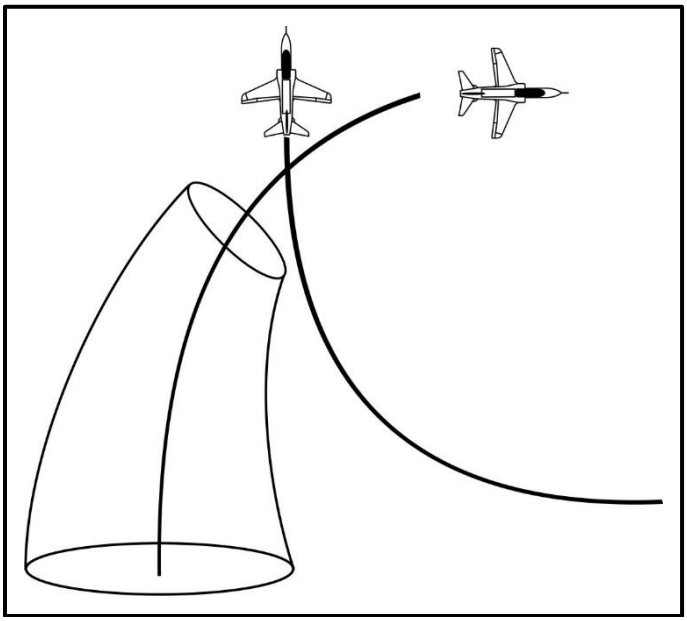

Figure 1-11 In-Close Overshoot

3. 3/9-line overshoot – an overshoot of the defender's 3/9-line that results in a positional role reversal (i.e., passing from behind to in front of the adversary).

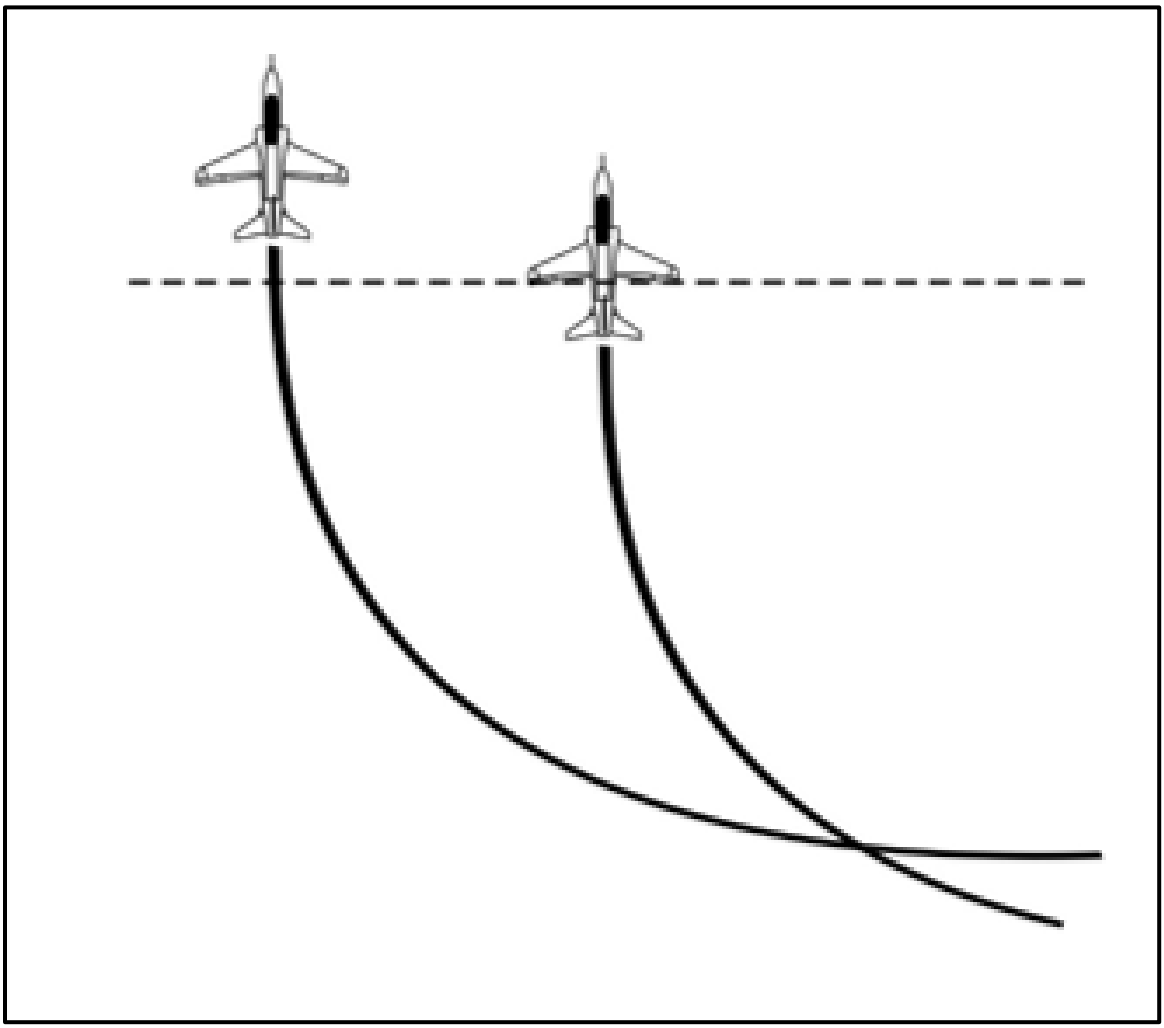

Figure 1-12 3/9-Line Overshoot

CNATRA Weapons Envelope

A weapons envelope is a graphical depiction of the limits of a particular weapon, centered on the target. A weapons envelope is at best a rule of thumb taking into account shooter and target airspeed, altitude, G, and weapon capability. Firing from within the envelope greatly increases the Probability of Kill (P_k). The CNATRA Weapons Envelope approximates both an infrared (IR) seeking (Fox-2) AIM-9 Sidewinder missile, and the 20 mm Vulcan cannon. Valid shots will require meeting appropriate wickets. For an IR Missile shot to be valid, it has to be taken with the defender in the HUD FOV, and the Shooter within the CNATRA Sidewinder Envelope. For a Tracking Gun Shot to be valid, it has to occur with the shooter in the Tracking Gun Envelope, and the trigger pulled with the pipper on the target. For a Snap Gun Shot to be valid, it has to occur with the shooter in the Snap Gun Envelope, trigger pulled early (>1 sec. prior) to establish bullets downrange at target distance, and the target must pass through the pipper.

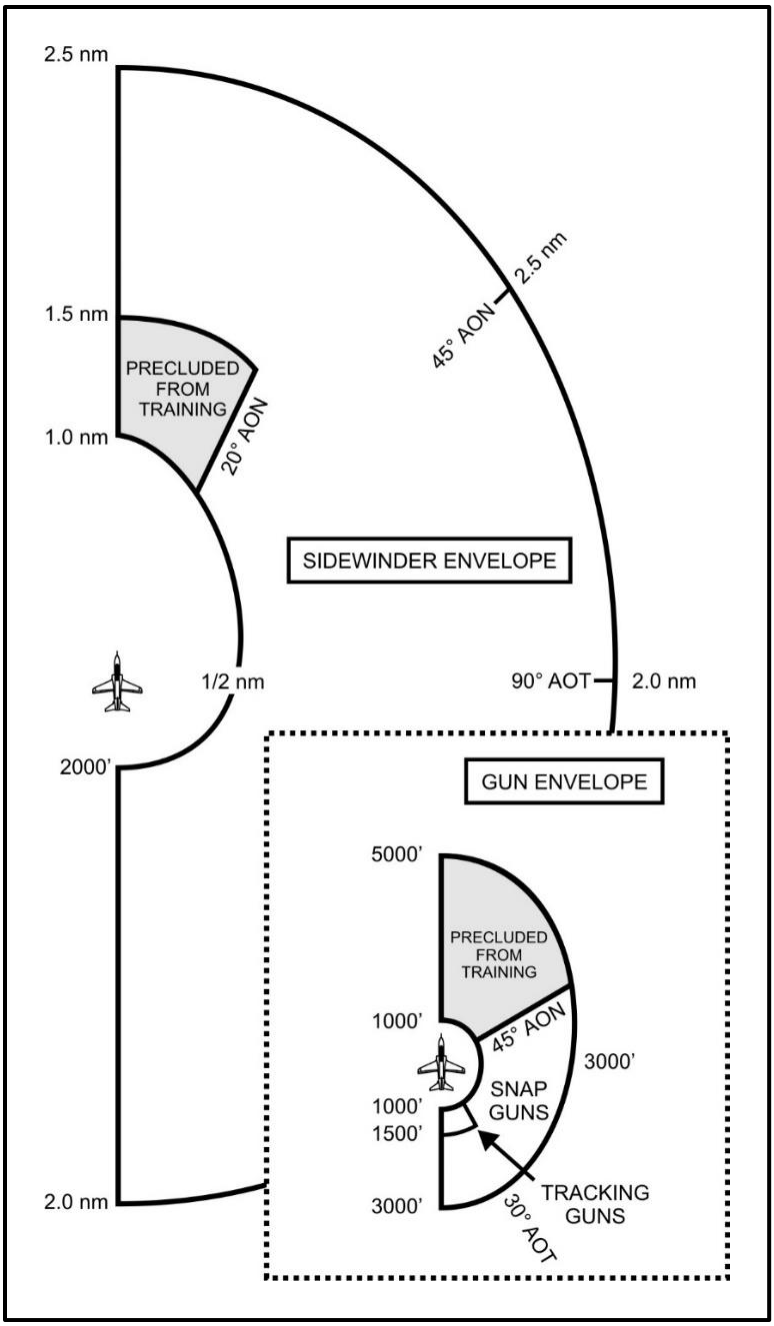

Figure 1-13 CNATRA Weapons Envelope

"FENCE(d) In/Out"

FENCE is an acronym for ***F***ire control, ***E***lectronic Countermeasures (ECM), ***N***avigation, ***C***ommunication, ***E***mitters. To FENCE(d) In means that administrative functions (to include G-warm) and proper setting of cockpit switchology have been completed prior to entering the combat area, and/or tactical portion of the flight. FENCE(d) Out means just the opposite, weapons switchology has been returned to a non-employment mode, and the fighters have returned to an administrative portion of the flight.

Yardstick

Yardstick is the comm brevity term for air-to-air (A/A) TACAN ranging. Yardstick gives DME ranging to the nearest aircraft transmitting 63 channels apart from the set TACAN channel. For example, 29X A/A gives ranging for an aircraft with 92X A/A. This is usually assigned in the brief but may be modified as required in flight. It will be displayed in the HUD if TACAN is boxed.

PADS

PADS is an acronym for the starting parameters of an engagement or maneuver: ***P***osition, ***A***ltitude, ***D***istance, ***S***peed.

OPS CHECKS (Fuel and G)

This is a call initiated by Lead and echoed by Wing after the 'Fenced In/Out' call and after every KIO stating the current fuel state and "good G" if within limits (fuel state in thousands of pounds, alibis by exception). If G is not within limits, say the actual max G. Do not delay ***aviating*** and ***navigating*** back to Spread just to ***communicate*** the 'fuel and G' call.

"Speed and Angels"

This is a call initiated by Lead and echoed by Wing when in position (on PADS) for the next maneuver set and it ***implies both fighters are visual***. Lead should initiate this call when Wing is in a good position. If not within +/- 100 ft, +/- 10 kts, and +/- .1 nm of PADS, call "standby." Reference the TACSOP for execution. The maneuver will not begin until all aircraft have called Speed and Angels.

"Knock It Off" (KIO)

This is a call made by any member of the flight to end an engaged maneuvering set (e.g., Gunsight Tracking). The KIO call is acknowledged by Lead, then by Wing. The KIO procedures are described in detail in the TACTICAL ADMINISTRATION section. ***Disciplined KIO comm and procedures are paramount for safety as the post-KIO portions of flights have historically been where most incidents occur.***

"Joker/Bingo" Fuel

Joker fuel is a pre-briefed fuel state above Bingo intended to advise Lead of the flight's fuel status.

Bingo fuel is the fuel state at which maneuvers must be terminated to rejoin the flight and comply with local course rules to the destination airfield in order to preserve sufficient fuel for approach(es) and landing(s). Be aware that this term also represents an emergency fuel state when briefed as such. Refer to local SOP regarding fuel management.

CHAPTER TWO
ADMIN

200. INTRODUCTION

ADMIN encompasses the non-tactical portions of the flight. For standard ADMIN procedures, reference the SOP/TACSOP and FORM FTI. The following paragraphs describe the aspects of ADMIN that are new to this stage of training.

201. PUSHING TO SPREAD

The flight lead will brief when to expect the 'push-out' signal to Combat Spread from Admin Cruise. Lead will give this signal by pushing their palm out and away towards the canopy. Respond with a head nod and add a sufficient amount of power while taking a cut away. Establish a 10° to 20° heading differential, set the nose attitude as required (if in a climb), and reset power as necessary to maintain 10 kts of excess airspeed for every 10° of heading differential while moving out on bearing. Hold this heading differential until achieving a lateral separation of 0.8 to 1.0 nm, then turn to parallel Lead. Anticipate bleeding the excess airspeed in the turn to parallel Lead's heading. As a visual cross-check, if the white or black letter on the tail of Lead is visible, the distance between Fighters is .7 nm or closer. If a character on the orange tail is visible but cannot tell exactly what it is, the distance is roughly .8 nm. If only an orange tail is visible with no white/black character, the Fighters are outside of .9 nm. These distances are environmentally dependent, so aircrew will need to develop an eyeball calibration by assessing the size of Lead and cross-checking with the A/A TACAN.

NOTE

> The sight picture from a good Combat Spread position will force the pilot to turn their head somewhat uncomfortably while looking down their shoulder.

Common errors:

1. Taking too aggressive a cut away, resulting in a sucked position.

2. Taking too little of a cut away for the excess airspeed, and either going acute or never getting out to the appropriate abeam distance.

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CHAPTER THREE
TACTICAL ADMINISTRATION (TAC ADMIN)

300. INTRODUCTION

TAC ADMIN encompasses everything that occurs administratively from FENCE-in to FENCE-out.

301. WEAPONS SYSTEMS

Review "NATOPS, PART VIII: WEAPONS SYSTEM, A/A MASTER MODE" for information regarding A/A modes and HUD symbology.

The most notable difference between the Navigation and Air-to-Air master modes is the lack of bank angle display. (When maneuvering in relation to other aircraft, specific bank angle is less important.) The selected sub-mode of Air-to-Air is indicated by Lead Angle Computing (LAC), or Real Time Gun Sight (RTGS), displayed just above GUN on the right center of the HUD. An "X" over GUN indicates that the Master Armament switch on the Weapons Selector Panel is set to SAFE. The "X" is removed when the Master Armament switch is set to ARM.

The wingspan setting entered (31 ft default for T-45) sets the diameter of the reticle in both LAC and RTGS. With the Bandit's wingspan set, the reticle can be used to estimate range. At 1,000 ft, the Bandit's wings fill the inside diameter of the reticle. The center of the A/A aiming reticle (The 'pipper') indicates the computed impact point. The reticle flashes if it reaches the edge of the HUD field of view, and any shots are invalid until the reticle is brought back into the field of view (steady, not flashing).

1. LAC. In LAC, the pipper indicates the impact point of the round at 1,000 ft, but the fighter needs to track the target for at least one second to get a valid aiming solution. LAC is used for a non-maneuvering or mildly maneuvering target.

2. RTGS. In RTGS, the pipper indicates the instantaneous impact point of the round at 1,000 ft. Since it will take 1/3 of a second for the round to travel 1,000 ft, the fighter needs to place the pipper ahead of the target's flight path with an offset distance that accounts for this 1/3 of a second time of flight. RTGS is used for a hard-maneuvering target and will be the mode used in TAC and BFM. The technique for using the HUD is basically a matter of experience. As the fighter maneuvers, the reticle will move in the HUD as it compensates for G and yaw. For this reason, the more smoothly the fighter maneuvers while tracking, the more accurate the aiming solution will be.

302. COMBAT CHECKLIST

All combat checks should be accomplished either on deck or en route to the area, short of selecting MASTER ARM switch to ARM for the Gunsight Tracking Exercise.

Combat checks for the T-45 are as follows:

Cockpit – Secure.

STORES page – A/A, GUN, RTGS all boxed.

Environmentals – Note.

MASTER ARM switch – ARM (for the Gunsight Tracking Exercise).

303. G-WARM

A G-Warm shall be executed prior to beginning any TAC FORM maneuvering. A G-Warm requires a minimum of 180° of turn. Normally, the flight will conduct two 90° turns in opposite directions such that the flight ends up on the original heading. The turn will be executed to achieve 4 Gs for the first 90°, and then a peak to 6 Gs, easing to 4 Gs, for the second 90° of turn. This allows for the flight to efficiently complete a G-Warm while transiting to the working area. The G-Warm will be initiated with the wingman co-altitude in Combat Spread. Reference the TACSOP for specific instructions on how the G-Warm will be executed.

The inside fighter will lose sight of the trail fighter during the G-Warm. A "blind" call is not required from the inside fighter in this case. ***If the trail fighter loses sight of the inside fighter, a "blind" call will be made and the trail fighter will execute all turns as directed.*** The trail fighter should strive to be at the inside fighter's dead 6 o'clock after the first turn.

It is essential that both aircraft strictly adhere to the airspeed and G requirements outlined in the TACSOP, as there is a potential for a mid-air if the trail fighter floats the second turn.

Post G-warm, the flight lead will direct the flight to set 330 KIAS. Select MRT to accelerate and reset the throttle to maintain approximately 1,900 to 2,000 PPH to maintain 330 KIAS. The flight lead will make the call in the following format:

Lead: *"(Call sign), set 330"*

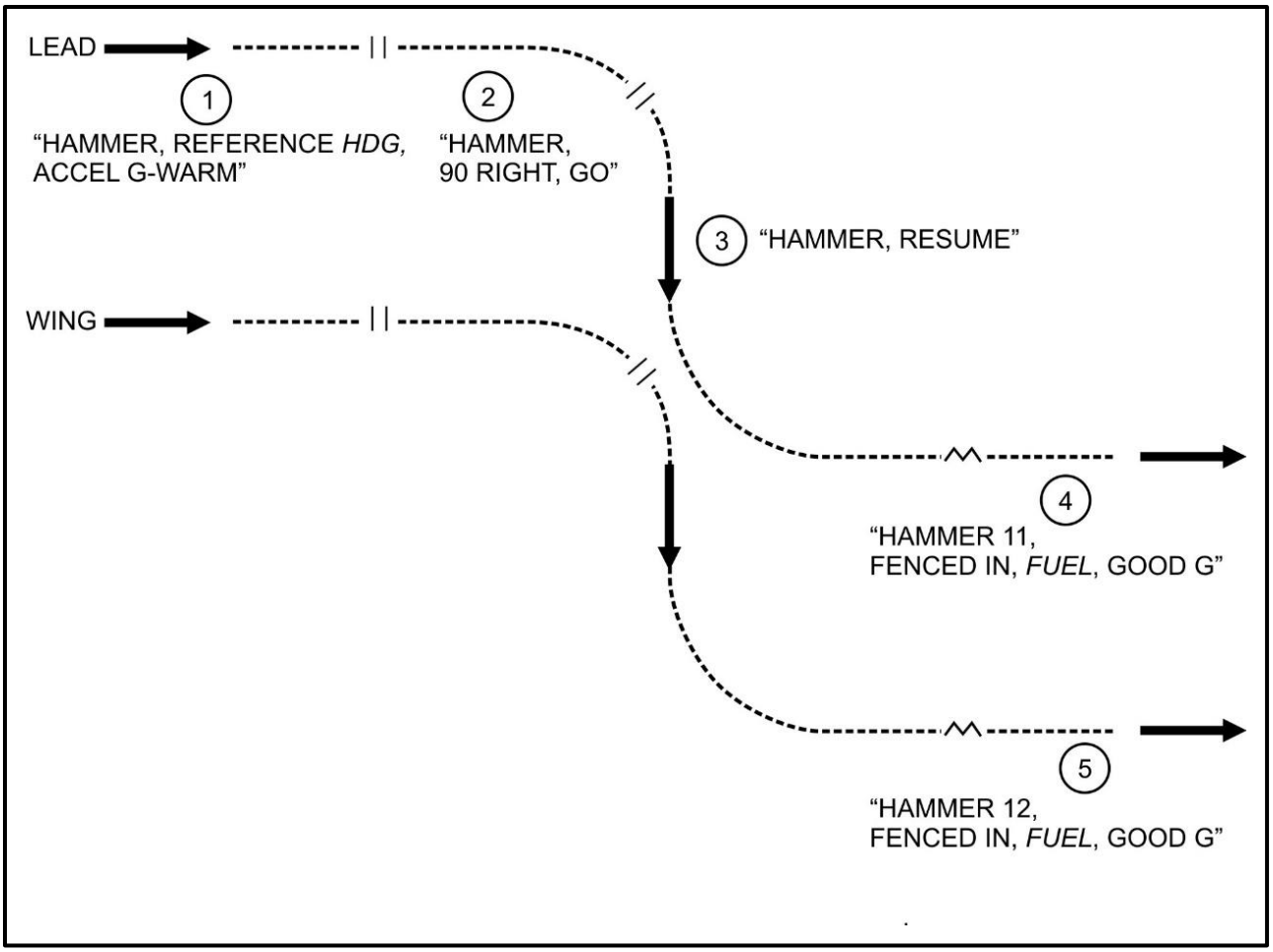

Figure 3-1 G-Warm (Into Wing)

Common errors:

1. Pulling for greater or less than 90° on the first turn, resulting in skewed geometry. This is more readily apparent when Wing does not end up at Lead's six o'clock during initial G-Warm away from Wing.

2. Poor Lift Vector and/or nose placement during the first turn and getting too fast (>400 KIAS) or too slow (<380 KIAS) prior to the second turn.

3. Not pulling right to 6 Gs in the second turn, or not maintaining at least 4 Gs thereafter.

4. Not getting the nose up quickly enough for the climb back to Spread, resulting in getting fast and going acute.

5. Post G-Warm – not intercepting Lead's bearing line and 300 KIAS.

304. FENCE IN

Most of the items in the Combat Checklist may be done at any time prior to fencing-in. The "fenced in" call indicates that all members are ready for the combat phase and have completed an assessment of themselves, the environment, their aircraft, weapons, and systems.

305. "KNOCK-IT-OFF" (KIO)

The Gunsight Tracking Exercise will be terminated with a "Knock-It-Off." It is very important to respect and understand the KIO, as it is a tool to ensure safety-of-flight. Thus, any member of the flight (even the Flight Surgeon in Lead's trunk) can call a KIO. Upon the KIO call, regardless of who called it, cease maneuvering, listen, and wait for Lead to initiate the follow-on comm sequence. Wing is responsible for flight path de-confliction unless blind and makes a 'blind' call. In this case, Lead, if visual, will have primary flight path de-confliction.
If Lead is also blind, they will call blind with altitude; for example, *"Hammer 11 blind*, ***altitude.***" Wing will echo this call*, "Hammer 12*, ***altitude***." Lead will dictate an altitude separation plan as applicable.

The priorities following the KIO are:

Stand up the throttle (if applicable), stop fighting, and maintain sight and/or SA throughout the maneuver. Post KIO cadence, use pull and power as required to the called flow heading to arrive in Combat Spread; sometimes this will be a 17-unit AOA pull at MRT, other times it will be different. During this process, ***avoid going belly up inside of a ½ mile from Lead***. If this is imminent, maintain sight of Lead and flow to the outside of the turn. Then, get back to bearing line while capturing 300 KIAS. Once on bearing, fix abeam distance and altitude. Lead's goal is to select a flow heading that puts Wing in Combat Spread. The flow heading may be adjusted after the initial turn. Once both fighters are visual, Lead will then initiate a '(Fuel) and (G)' call. Again, the initial 'Knock-It-Off, Knock-It-Off' call can be made by anyone. Reference the TACSOP for execution.

Common Errors:

1. Delaying the turn to the called heading.

2. Not adjusting pull and/or power to arrive in Combat Spread in a timely manner.

3. Being well below assigned Combat Spread altitude, and out of position following the KIO, yet pulling power prior to initiating corrections back to Spread.

4. Going 'belly up' to Lead inside ½ mile.

306. FENCE OUT

Post KIO, it is Wing's responsibility to maneuver back to Combat Spread. At some point, Lead will initiate the fence out. Set the MASTER ARM switch to SAFE, VCR tapes – as required, TACAN – mode as briefed, and select HUD – NAV master mode. Reference the TACSOP for execution.

307. REJOIN

The Rejoin gets the Fighters back into Parade formation prior to returning to base. It will be executed from Combat Spread and initiated at the "fence out" call from Lead. If in a position other than Combat Spread at the "fence out" call, utilize Tac Form corrections to establish Combat Spread position (sucked positions do not require corrections as join up may be executed from this sight picture). If beginning a tactical rejoin from an acute position, avoid closure into Lead until on or aft of Lead's 3/9 line. Expect Lead to be at 80% N2 during the rejoin, conditions permitting. It is critical to maintain sight of Lead at all times during the join. ***Within a half mile, never go 'belly up' for safety of flight.***

The ultimate goal of the Rejoin is to safely maneuver your aircraft to a position in which you can execute a join you have been taught previously. To execute the rejoin from Combat Spread, turn to establish a cut into Lead and descend. Modulate throttle position as appropriate for the unknown airspeed rendezvous. As nose to tail separation is established, maneuver the aircraft to a position in which a running rendezvous or CV rendezvous can be executed. Tactical rejoins from sucked positions require a small cut into Lead due to nose to tail separation already established. During these maneuvers, be co-altitude to slightly below Lead to avoid going 'belly up' inside of a half mile and losing sight of Lead as you turn to parallel Lead's heading. ***Sight of Lead must be maintained.***

If Wing fails to execute the basics during the running rendezvous, an overrun may occur. An overrun occurs when the wingman flies past bearing line, ending up acute. If Wing is approaching Lead's bearing line too quickly, speed brakes may be required to capture bearing line. If unable to stop on bearing line, maintain approximately 250-300 ft abeam Lead, decelerate, and work back to bearing line and once airspeed control is established, continue the join up. If in a CV rendezvous situation, execute the fundamentals (Altitude, Bearing and Closure), and use speed brakes if necessary to avoid an underrun.

Once Parade is established, the Rejoin is complete.

308. BATTLE DAMAGE CHECKS

The flight lead will initiate battle damage checks via the “cocked gun” hand signal. The wingman will maintain safe separation from Lead, elevate to check the top, descend and cross under checking the belly, and elevate on the other side checking the top again, remaining on that side. The wingman shall pass a thumbs up or communicate any problems. The flight lead will pass the lead to the wingman, the wingman will give the “cocked gun” hand signal, lead will then execute the battle damage checks, and be give the lead back upon completion. The flight lead will maintain responsibility for ATC/Safety of Flight frequency communication throughout the battle damage checks.

CHAPTER FOUR
BASIC POSITION-KEEPING AND CORRECTIONS

400. INTRODUCTION

The following section gives basic procedures for maintaining and regaining position. These are not specific maneuvers to perform, but rather tools to reference while performing various maneuvers or correcting for deviations that arise while flying TAC FORM.

401. MAINTAINING POSITION

The laws of physics that affect position-keeping when flying in Combat Spread are the same laws that affect position-keeping while flying in Parade. The only major difference is the pilot's ability to discern deviations due to the relatively large separation between the fighters. A disciplined inside/outside scan is crucial to seeing deviations, but ***the real key to staying in position is basic air work***; how long do you think Wing could stay in Parade position with five more knots than Lead, or two degrees difference in heading? While Combat Spread is much farther from Lead than is Parade, the same principles of position-keeping apply.

Scanning fuel flow as much as airspeed will pay dividends in keeping airspeed under control while attempting to maintain proper TAC FORM positioning. 1,900 to 2,000 PPH is a good reference point for maintaining 330 KIAS in level flight. For heading control, listen to the flow heading called by Lead. Request Lead to "say flow" if unsure of the heading called. Altitude and changes in altitude are fairly easy to see, either with the altimeter or by comparing Lead's jet to the horizon.

402. POSITION CORRECTIONS

There are three variables the wingman must solve for to correct and maintain position in Combat Spread. TAC FORM positioning order of priorities is:

1. Bearing line.

2. Abeam distance.

3. Altitude.

A rapid inside-outside scan will be required to maintain the proper bearing, abeam distance, and altitude. The right combination of airspeed, heading, and power will control the proper Combat Spread position, so in the most basic sense: airspeed controls bearing, heading controls distance, and nose attitude controls altitude. Any deviation in just one of these parameters will cause a deviation to the accepted Combat Spread position. It is possible, and preferred, to apply multiple corrections simultaneously to regain or maintain Combat Spread position.

A few rules to corrections while trying to fix the above variables:

1. ***Always lead acute or altitude corrections***. Wing should be accelerating back to 330 KIAS before arriving on bearing line while fixing an acute, and/or starting level-offs before blowing through altitudes. Otherwise, Wing will quickly become sucked.

2. ***Never lead sucked or abeam distance corrections***. Wing should be on bearing line when correcting from a sucked position before climbing to decelerate. Wing should also hold a heading correction until reaching the appropriate abeam distance prior to turning to parallel.

Now is a good time to introduce the concept of ***energy conversion***. Getting proficient at trading altitude for airspeed, and vice versa, will pay dividends in making smooth, expeditious, and efficient corrections while attempting to maintain Combat Spread.

Maneuvers and corrections need to be timely and smooth. It is important to note that erratic and/or abrupt control inputs will just aggravate all involved. Learn to make smooth and controlled, yet aggressive corrections; sometimes a 7-unit AOA pushover will be required. Other times, rolling inverted and pulling will achieve the same result. Going up on a wing might also be used if applicable. Whichever technique is used, Wing should be smooth in making corrections.

403. CORRECTING FOR A SUCKED POSITION

For ***slightly sucked*** positions (10° or less), simply add power, accelerate above 330 KIAS, and reset power to maintain 330 KIAS once on bearing.

Correcting for a ***fully sucked*** position (greater than 10°) in the T-45, is a matter of trading altitude (potential energy) for airspeed (kinetic energy) to start the correction and reversing the trade to stop the correction. Simply push the nose over to gain airspeed. Take the extra airspeed once level and drive forward to the bearing line. When on bearing line (never lead a sucked), smoothly and aggressively pull the nose up, and climb back to the desired altitude. This significant pull up and nose-high attitude will rapidly gain altitude, while concurrently decelerating. As a useful rule of thumb, every 1,000 ft of altitude gained or lost in a climb or descent will lose or gain roughly 30 knots respectively. This correction can be accomplished with little to no throttle modulation, but for a large deviation power may need to be added to further expedite the correction. Adjust the climb attitude to arrive back on bearing, at altitude, and 330 KIAS with about 2,000 PPH fuel flow. This is simplified, but the premise works for all variations of sucked corrections as long as the magnitude of the correction matches the magnitude of the deviation.

404. CORRECTING FOR AN ACUTE POSITION

For ***slightly acute*** positions (10° or less), simply reduce power, decelerate to below 330 KIAS, and reset power to maintain 330 KIAS just prior to arriving on bearing.

For ***full acute*** positions (greater than 10°), make level turns or climb and decelerate. Climbing and decelerating can be more fuel efficient. Simply climb to trade airspeed for altitude and start a drift aft towards bearing line. Before arriving on bearing (always lead an acute) push the nose over and descend back to altitude. Adjust power as necessary to arrive on bearing with 330 KIAS. As in the sucked correction, this method can be utilized with little to no throttle modulation, but for a large deviation power modulation may be required to expedite the correction.

Another option to get back to bearing line when acute is to couple turns with an altitude excursion and power adjustment. Use nose position and power as required to arrest downrange travel. This might involve greater pure level deviations or by combining turns with power and altitude adjustments. ***It is very important to make corrections smooth and precise. Abrupt and erratic turns will only result in a loss of SA.***

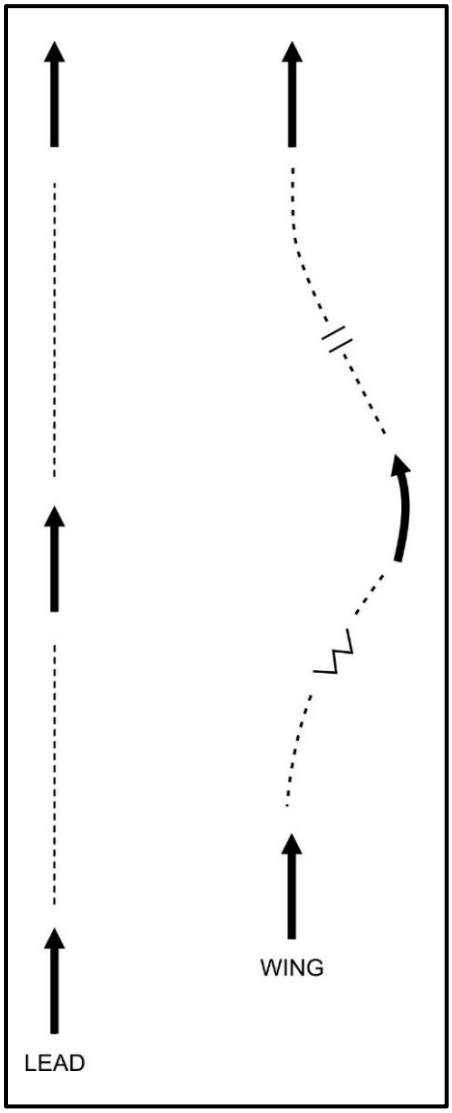

Figure 4-1 Acute Correction

405. ABEAM DISTANCE CORRECTIONS

Abeam distance corrections are simple, add a few knots of airspeed and take a cut into or away from Lead. As a rule of thumb, add as many knots as degrees of heading change.

406. ALTITUDE CORRECTIONS

With all other variables solved, add power and set nose attitude to maintain slightly greater than 330 KIAS for a climb. The greater the nose attitude, the more excess airspeed is required. For a descent, pull power and/or partially extend the speed brakes so as not to accelerate excessively and become acute.

407. COMBINING CORRECTIONS

Corrections for bearing may be combined with corrections for abeam distance. For example, if acute and tight, a check turn away will have the net result of increasing abeam distance, stopping downrange travel, and moving back to bearing. If acute and wide, a check turn in to Lead will net the same result. If sucked and tight, combining a level acceleration or trade of altitude for airspeed and a check turn away from Lead will result in moving up on bearing and increased abeam distance. If sucked and wide, a level acceleration or trade of altitude for airspeed and a check turn into Lead will net the same result. Regardless of the technique used, prioritize capturing (in order) bearing line, abeam distance, and altitude.

CHAPTER FIVE
TAC FORM PROCEDURES

501. MECHANICS

Maximum Rated Thrust (MRT), G to maintain airspeed mechanics will be used for all maneuvers. Select MRT, unload, roll, and pull to maintain the briefed airspeed (300 or 330 KIAS). Once the desired airspeed is established in the pull, maintain G in the seat by feel. Prioritize looking outside to reference positioning and checkpoints throughout the maneuver. After rolling out of the turn, reset the throttle to maintain the briefed airspeed.

502. COMMUNICATIONS

The flight lead will call maneuvers in the following format:

Lead: *"(Call sign), (maneuver)"*
Lead: *"(Call sign), flow (heading)"*

These calls can be combined in to one transmission.

When Lead initiates the call, look at the HSI to conceptualize the maneuver being called, and listen for the flow heading. ***The command of execution is the flow heading***. However, in the fleet, the flight lead will call a maneuver, initiate the maneuver, and the flow heading will be called mid-maneuver or adjusted after the maneuver is complete to fix positioning or as dictated by the tactics being employed.

503. CHECK TURNS

A Check Turn is typically used to turn the section 30° or less. It can be used to help the wingman fix positioning, accomplish specific training objectives (by forcing the wingman out of position), or facilitate a Tac Admin set-up (see Gunsight Tracking Exercise). Simply roll in to a 30° AOB turn, maintain airspeed, and turn to the flow heading. Check turns can be initiated in one of two ways:

Lead: *"Hammer, Check (left/right)"*
Lead: *"Hammer, flow 360"*

Or simply with the flow heading:

Lead: *"Hammer, flow 360"*

504. SHACKLE TURNS

Shackles allow Lead to re-dress the section (get Wing back into position), or to have the fighters maneuver to opposite sides of the formation. If Lead desires to keep the section on the current heading, both aircraft begin a level, MRT, G to maintain airspeed turn toward each other for

approximately 45° of heading change. Both aircraft will roll out with the flow heading on a 45° benchmark and wait for flight path passage while resetting the throttle to maintain 330 KIAS. The wingman is responsible for altitude de-confliction and should keep Lead in sight throughout the maneuver. If both aircraft cross over each other and reverse back to the flow (original) heading ***at the same time*** (roughly 1.5 to 2 sec after the pass), both should come out of the turn on bearing line at the appropriate abeam distance.

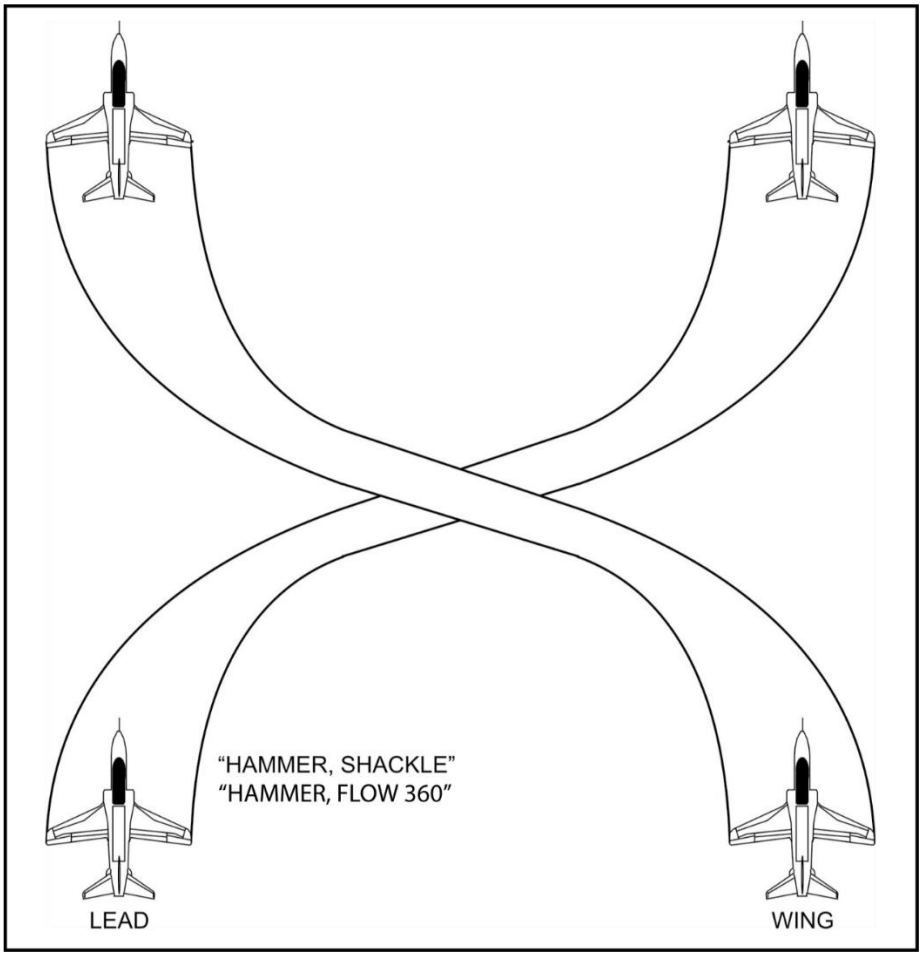

Figure 5-1 Shackle

Common Errors:

1. When in position and Shackling straight ahead, Wing turns all the way to the pass (greater than 45°), instead of rolling out on the 45° benchmark.

2. Not maintaining MRT, G to maintain in either turn.

3. Not scanning airspeed and getting either too fast or too slow.

4. Not making the cross happen.

5-2 TAC FORM PROCEDURES

505. SHACKLE EXERCISE

A Shackle can be used to help the wingman regain bearing line from either an acute or sucked position. The Shackle Exercise will be used to practice these corrections. The exercise can be initiated with the wingman either in position or out of position (acute or sucked). If in position, Lead will check the flight 30° into or away to force the Wing out of position.

If acute, turn more than 45° to make the cross happen. If excessively acute, turn up to 90° to make the cross happen. If sucked, turn for less than 45° to make the cross happen. If excessively sucked, no turn at all may be required to make the cross happen. In all cases, and regardless of the initial deviation and correction used to make the cross, the wingman must look outside after the cross and adjust the reversal timing and pull to arrive on bearing. Ultimately, a shackle is a cooperative maneuver requiring both fighters to make the cross happen, adjust reversal timing for the desired abeam distance, and modulate their pulls to arrive on bearing.

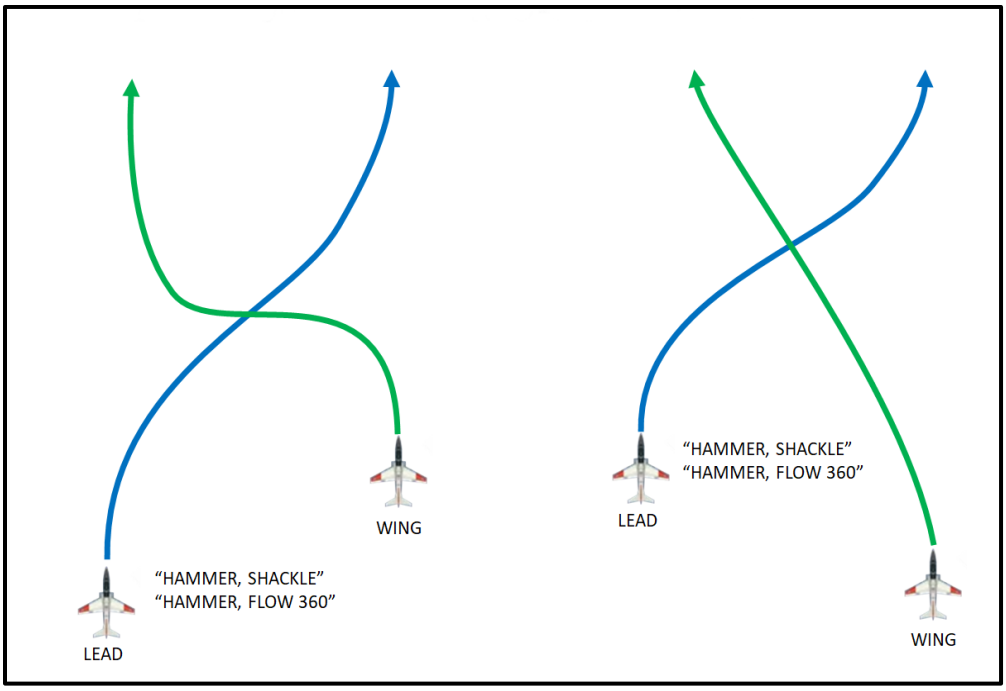

Figure 5-2 Shackle Exercise

506. TAC TURNS

The Tac Turn is an engaging turn in which flow heading changes by 90°, and the Fighters swap sides of the formation. A Tac Turn allows the section to maneuver rapidly to engage a threat in the beam. Tac Turns will be performed at MRT and G to maintain 330 KIAS. At the completion of the turn, the wingman will be back in Combat Spread on the opposite side of the formation at 330 KIAS. The turn begins immediately following Lead's comm:

Lead: *"Hammer, Tac left"*
Lead: *"Hammer, flow 090"*

507. TAC TURNS INTO WING

Initially Wing will flow straight ahead and do nothing as Lead starts the turn into. Begin the turn in the called direction just prior to being able to look down Lead's intakes. About halfway through the turn Wing should pick up Lead over the shoulder as Lead rolls wings level. Expect most of Lead's turns to be into Wing as this allows the wingman to get the most training on timing and corrections. If the wingman is out of position to start the maneuver, an adjustment to maneuver timing will be required to come out of the turn in position.

If ***acute***, turning when Lead's intakes are approaching will result in being acute. Wing should wait slightly longer to begin the turn; this will help fix bearing line (first priority) out of the turn but will induce a wider abeam distance. To fix abeam distance, simply put in a correction to fix being wide coming out of the turn.

If ***sucked***, turning when Lead's intakes are approaching will result in being sucked. Wing should start the turn slightly sooner in order to arrive on bearing line at the completion of the maneuver. This will induce a tighter abeam distance and a correction can be made out of the turn.

Key to Success: Do not solely rely on timing. With consistent mechanics, work to regain sight of Lead as soon as possible so a proactive correction can be made prior to rolling out of the maneuver prioritizing (in order) bearing, abeam distance, and altitude.

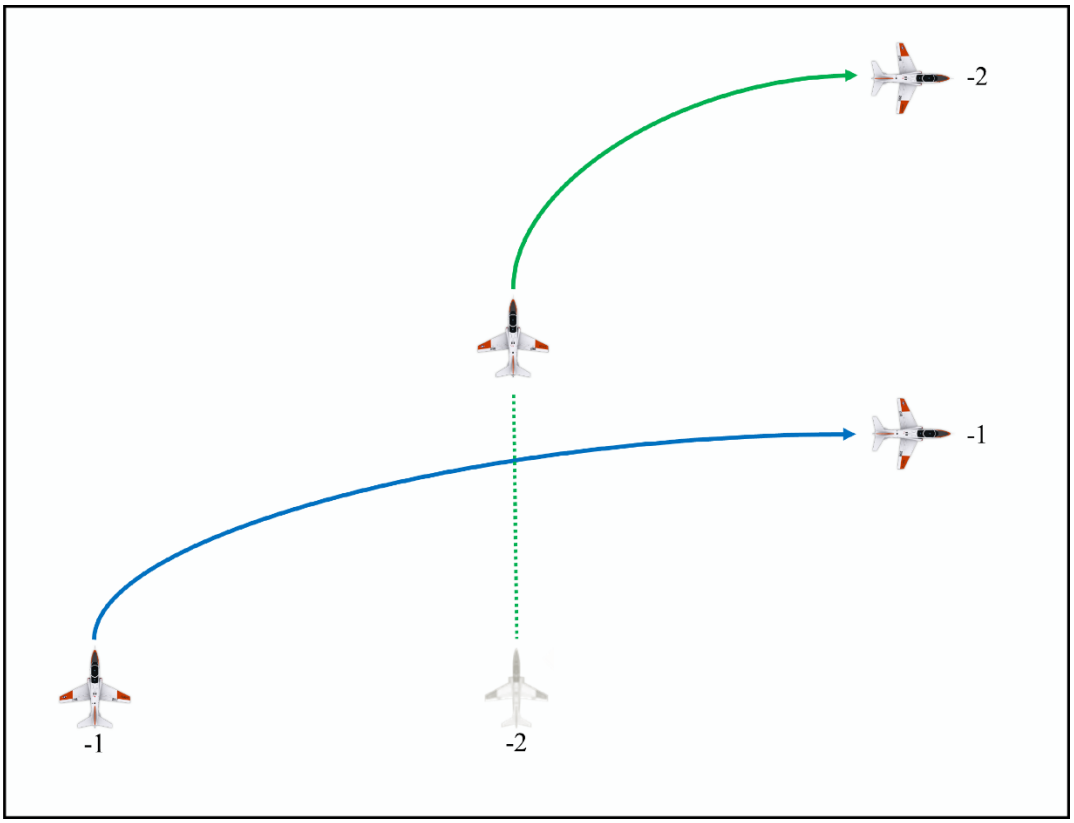

Figure 5-3 Tac Turn

508. TAC TURNS AWAY FROM WING

Immediately begin a turn into Lead for 90° using MRT, G to maintain 330 KIAS. Lead will maintain heading waiting for the correct sight picture and will start the turn as appropriate. Wing should arrive at the new heading, on bearing, as Lead is about halfway through the turn.

509. IN-PLACE TURNS

The In-Place Turn is an engaging turn that changes flow heading by 180° and causes the Fighters to swap sides of the formation. Both Fighters should begin and finish the turn at the same time. An In-Place Turn is used to engage a bandit that is in the section's rear quarter, in the direction of the turn. The turn begins immediately following Lead's comm:

Lead: *"Hammer, In-Place right"*
Lead: *"Hammer, flow 180"*

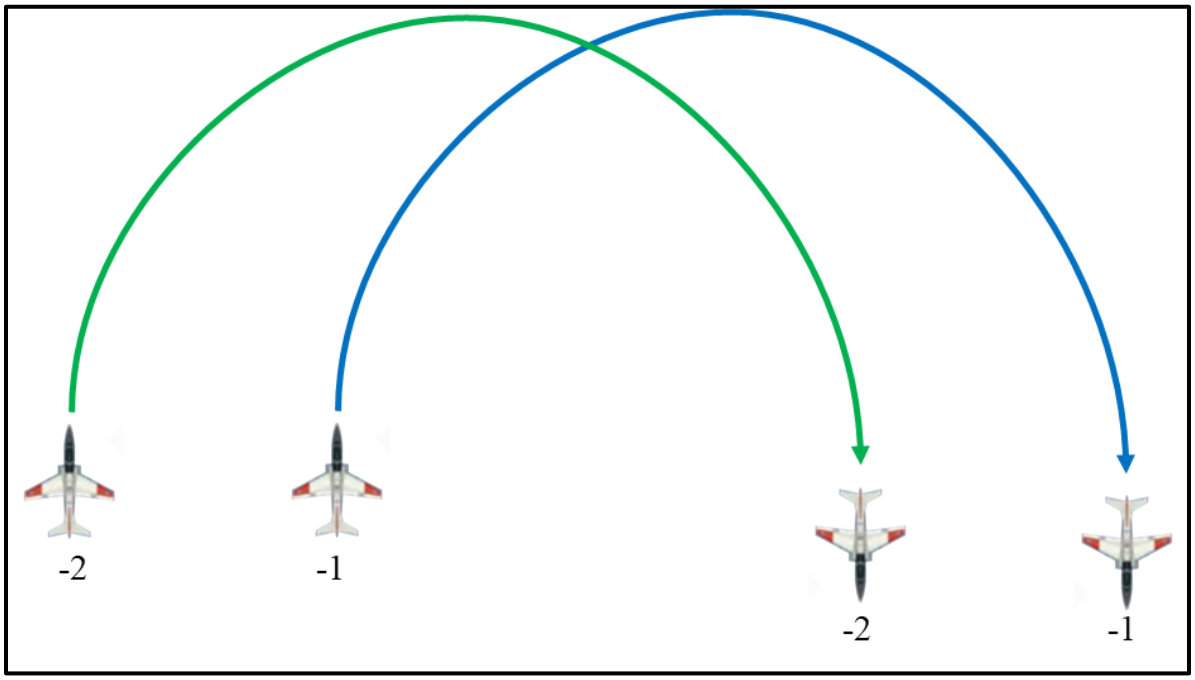

Figure 5-4 In-place Turn

Common Errors:

1. Slow initiation of the maneuver.

2. Not assessing corrections needed during the turn.

3. Not maintaining altitude while blind in the turn.

510. IN-PLACE TURNS INTO WING

Once the flow heading is called, immediately turn away from Lead using MRT, G to maintain 330 KIAS. After roughly 135° of turn, look for Lead at the 5 or 7 o'clock position, and assess bearing line. Lead will strive to fix Wing's position errors during the turn but this will require Wing to have consistent mechanics. Since In-Place Turns take twice as long to accomplish as Tac Turns, poor basic air-work from Wing will result in amplified deviations after the turn is complete.

511. IN-PLACE TURNS AWAY FROM WING

Once the flow heading is called, immediately turn into Lead using MRT, G to maintain 330 KIAS. The first check point in the turn will be after 90° of heading change. The wingman should be able to sight down Lead's tailpipe at the 90° checkpoint. If behind (nose in lag), the wingman will need to "wrap up" the turn (increase turn rate) using a slight overbank to descend and increase G available to arrive on bearing. If ahead (nose in lead), momentarily ease the pull to stop downrange travel, and reset the pull. Of note, wrapping up or easing a pull will induce abeam distance errors. Anticipate abeam distance errors and start planning a correction out of the turn to correct. ***As a safety note, descending out of altitude during an In-Place Turn away from Wing should only be done with situational awareness to Lead's position. When co-altitude, be especially careful not to pull an excessive amount of lead, ease excessively to correct, and significantly collapse the range. This could result in the outside fighter being belly up inside of 0.5 nm.***

512. IN-PLACE TURN EXERCISE

The In-place Turn Exercise will be used to practice the corrections for acute or sucked positioning at the start of an in-place turn. The exercise can be initiated with the wingman either in position or out of position (acute or sucked). If in position, Lead will check the flight 30° into or away to force Wing out of position. All in-place turns for this exercise will be executed AWAY from the wingman.

If acute, overbank, select MRT, pull to maintain G for 330 KIAS, and descend to arrive at the 90° checkpoint directly behind Lead. The descent will provide more G available to increase turn rate to arrive at the 90° checkpoint. If acute and airspeed greater than 330 KIAS, pull to bleed excess airspeed and descend as required to arrive in position at the 90° checkpoint. If acute and airspeed excess is excessive (greater than 350 KIAS), throttle modulation may be required to intercept 330 KIAS in the pull and an overbank will be required to arrive in position at the 90° checkpoint. Once there, reset the nose attitude, select MRT, and pull to maintain 330 KIAS in the turn.

If sucked, delay the turn slightly to arrive in position at the 90° checkpoint. If sucked and excessively fast, less of a delay will be required. Pull to bleed airspeed to 330 KIAS and arrive in position at the 90° checkpoint.

In any case (acute or sucked), and if excessively slow (less than 310 KIAS), lower the nose to gain airspeed, re-establish 330 KIAS in the pull, and apply the corrections described above to arrive at the 90° checkpoint.

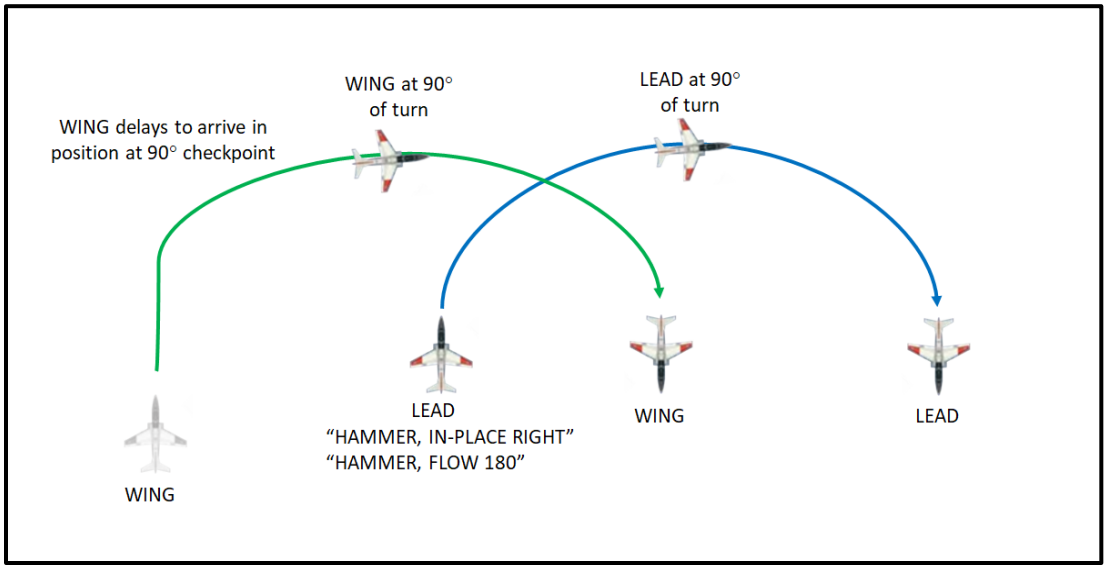

Figure 5-5 In-Place Turn Exercise (from sucked position)

513. CROSS TURNS

The Cross Turn is an engaging turn where both fighters cross flight paths and change flow heading by 180°. Out of Cross Turns, the Fighters are on the same relative side of the formation; additional comm is needed for safety of flight de-confliction because the Fighters are going to make a relatively close pass. ***This is the student's first introduction to a high-aspect pass, the echoing of a pass, and observance of a 500' safety bubble***.

Lead will initiate the Cross Turn, "Hammer, Cross Turn, Hammer 11 low." In early TAC FORM flights, Wing will usually be the high Fighter, so the response will be, "Hammer 12 high." As Wing transmits the response, both fighters will select MRT and begin rolling to execute a pull to maintain 330 KIAS to pull directly over top each other. Both Fighters will end up turning 180° to the new flow heading. The pass will occur just prior to either jet achieving 90° of turn. Wing should be right above and slightly inside of Lead's turn radius. The degree to which Wing goes nose-low is predicated on being ***no closer than 500 ft*** above Lead at the pass.

Once the pass is safely assured, Wing should overbank briefly, maintaining the pull for 330 KIAS. The section started out at 0.8 to 1.0 nm abeam prior to the Cross Turn, but the wingman end up on bearing wider than usual (greater than 2.0 nm abeam) due to turn radius at MRT, G to maintain 330 KIAS. Wing can gain excess airspeed, take a cut in to Lead once on bearing, and simultaneously climb back to altitude and 330 KIAS. If on bearing, approximately 30 KIAS excess airspeed will be required to maintain bearing during a 1,000 ft climb.

If Wing is blind out of the Cross Turn, only turn 180° to parallel the new flow heading ***but do not climb***. Once visual, proceed with correcting back to the proper abeam distance and altitude. This assumes that Wing is on bearing after 180° of turn. Typically, Wing ends up slightly sucked, and will also have to drive to bearing line after correcting for abeam distance. Once Wing hits the bearing line with proper abeam distance, climb back up to altitude. Should Wing show up acute after 180° of turn, assuming visual, Wing will need to incorporate an acute fix while collapsing the abeam distance.

NOTE

Leads may brief to "set 300" prior to initiation of the cross turn if the intent is to set up for co-altitude tactical formation maneuvering or the gunsight tracking exercise. The mechanics are identical and will result in the wingman being at approximately 1.5nm abeam with airspeed greater than 300 KIAS.

Common Errors:

1. Not talking while turning at the start, resulting in Wing being outside Lead's turn.

2. Not selecting MRT early enough or maintaining G for 330 KIAS throughout the turn.

3. Overbanking too much and/or too early, resulting in a close pass with Lead.

4. Going belly-up to Lead to correct heading/abeam distance, while climbing through Lead's altitude.

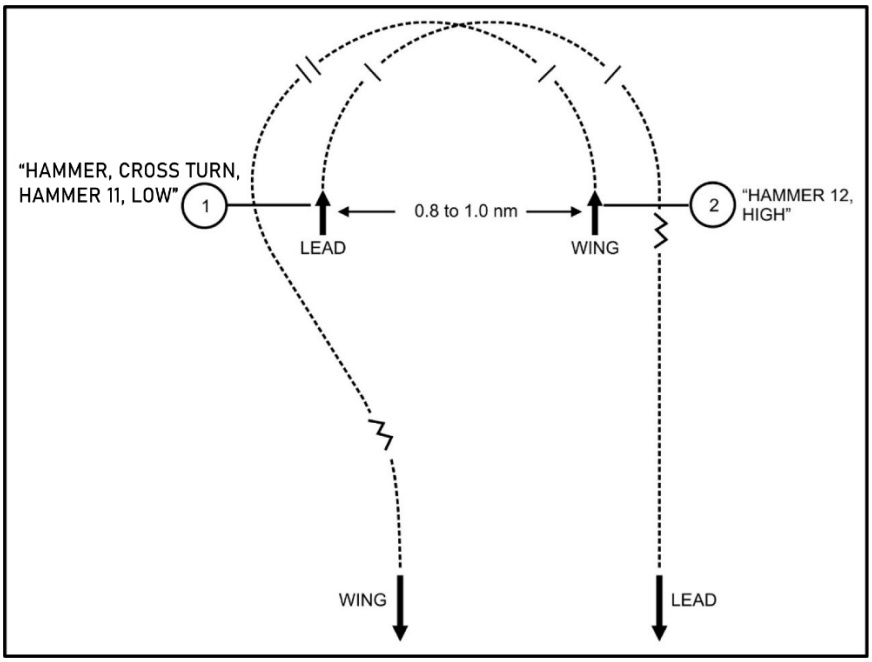

Figure 5-6 Cross Turn

CHAPTER SIX
ADVANCED TAC FORM PROCEDURES AND MANEUVERING

601. ADVANCED TAC FORM MANEUVERING

Some turns will not be for 90° or 180° of heading change. Even when called turns conform to these benchmarks, Lead may adjust the flow heading to fix positioning, manage the area, or simply maneuver the section. For any of the turns Lead may elect to maneuver within +/- 30° of standard parameter heading changes.

Tac Turns to 60° or 120° from the original heading. A Tac Turn into Wing will force Wing to adjust maneuver timing (before or after Lead's intakes approaching). Use the HSI to determine the outcome. If less than 90° degrees of turn to the flow heading, then the wingman will need to turn late. If more than 90° degrees of turn to the flow heading, then the wingman will need to turn early.

- ***Tac Turns into Wing <90°*** will result in Wing being well-acute out of the turn if no adjustment is made. Wing must begin the turn later than normal. For a typical Tac 60, begin the turn approximately 1 second after Lead rolls out and disappears beyond the aft visual limit. Maintain consistent mechanics, regain sight as soon as possible in the turn, and make a proactive correction to maintain/regain bearing line prior to rolling of the turn.

- ***Tac Turns into Wing >90°*** will result in Wing being well-sucked out of the turn if no adjustment is made. Wing must begin the turn earlier than normal. Maintain consistent mechanics, regain sight as soon as possible in the turn, and make a proactive correction to maintain/regain bearing line prior to rolling of the turn.

602. CO-ALTITUDE TACTICAL FORMATION MANEUVERING

During the TAC stage, the concept of co-altitude Tac Form Maneuvering will be introduced. This will require different eyeball calibration and timing to execute turns effectively. This exposure will prepare the wingman to set specific PADS prior to the Gunsight Tracking Exercise in this stage as well as future stages of training such as BFM and FTX. It is important to note that mechanics and timing must be adjusted when executing turns at different airspeeds and abeam distances. For example, when executing Tac Form at 300, 330, or 350 KIAS, turns will be conducted at MRT utilizing G to maintain airspeed, and while executing Tac Form at 200 KIAS, turn mechanics will target 17 units AOA with power modulated to maintain airspeed.

During the TAC stage, co-altitude maneuvering will be executed at 1.5 nm abeam Lead at 300 KIAS and will consist of both on- and off-heading turns. Prior to executing turns, Lead, will make a call to "set 300." This can also be accomplished by setting 300 KIAS prior to a cross turn. Differences are as follows:

1. ***Shackles***. Co-altitude shackles will require a pass to be called ("high" or "low" and are typically initiated by Lead) in order to preserve a 500 ft bubble around both aircraft. This will

require one fighter to climb slightly and the other to descend. Shackles to 1.5 nm will require longer timing after the pass before reversing to the flow heading. Methods for making corrections remain the same.

2. ***In-Place Turns***. There is no difference for the In-Place Turns.

3. ***Tac Turns***. Tac Turns into Wing require slightly different timing. Instead of waiting to look down Lead's intakes, Wing should wait slightly longer (until Lead reaches the 5 or 7 o'clock position) before starting a MRT, G to maintain airspeed turn. A Tac Turn initiated with the wingman significantly out of position may require a pass to be called ("high" or "low" typically initiated by Lead) and will require both aircraft to maneuver cooperatively to preserve a 500 ft bubble around both aircraft.

603. GUNSIGHT TRACKING EXERCISE

The Gunsight Tracking (GST) exercise introduces the basic Tac Admin setup for perch BFM. The concepts of the Bubble, Attack Window, and Control Zone will be introduced as well as methods for employing tracking gun shots. A thorough knowledge of the Training Command Sidewinder Missile and Gun envelopes (CNATRA weapon envelopes) will be required to successfully complete this exercise. The student will be exposed to the process by which a fighter can control range, angles, and closure to effectively employ a particular weapon system against a target. In keeping with traditional air-to-air training terms, the terms 'Attacker' and 'Defender' will be used in the following discussion. Lead will be representing the 'Defender' and Wing will be representing the 'Attacker.' Lead will be maneuvering cooperatively to allow Wing to achieve a tracking solution fairly quickly (180° to 270° of turn), yet hard enough during the initial Attack Window Entry to simulate a realistic scenario. CNATRA ACM Training Rules will be briefed for flights intending to execute GST. If any of the training rule requirements are not met, GST will be terminated after 180° of turn.

The goal of the GST is to recognize the Attack Window, execute Attack Window Entry Mechanics to arrive in the Control Zone with range, angles, and closure under control, and to employ follow-on tracking gun shots on the Bandit. Lead pursuit will be used to employ shots, and lag to stay offensive and manage the forward portion of the Control Zone. Tracking gun shots will be employed just forward of the Control Zone at the Defender's 0-30° AOT cone, at a distance of 1,000 to 1,500 ft. ***During the GST maneuver, the throttle will be advanced to MRT at the "Check L/R 50" call and kept there through the AWE execution for compressor stall mitigation***. After the AWE, with stabilized AOA, the throttle may be modulated as required by the Attacker as to maintain closure in the Control Zone, and to manage pipper placement for a tracking gun shot. A High Yo-Yo maneuver may be used to slow or stop closure, and/or increase nose-to-tail separation, or a Low Yo-Yo maneuver may be used to decrease nose-to-tail separation, as applicable.

604. GST ENTRY AND EXECUTION

The set up for entry into the GST exercise will involve some three-dimensional geometry and communication to get both Lead and Wing at 16,000 ft, 350 KIAS, and 1.5 nm abeam. This procedure is listed below:

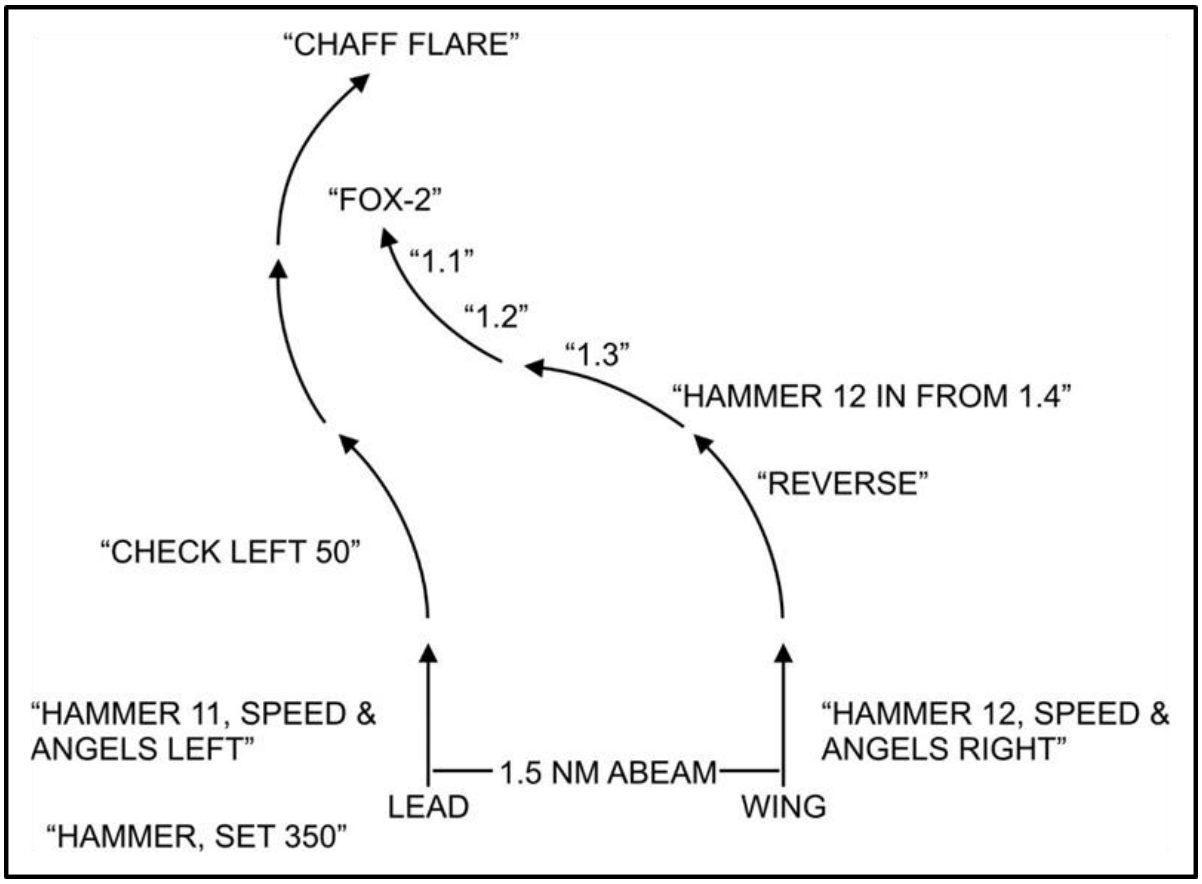

Figure 6-1 Set-up for GST

The set-up will be initiated by Lead:

Lead: *"Hammer, setting up for GST"*

Both Lead and Wing maneuver to achieve 16,000 ft, 300 KIAS, and 1.5 nm abeam. Once both aircraft are established there:

Lead: *"Hammer, set 350, check tapes, master arm"*

Both aircraft will select MRT, place the master arm switch to the ARM position, and execute a level acceleration, resetting power as required to establish 350 KIAS. The master arm switch will remain in the ARM position until directed to fence out. Lead will initiate the Speed and Angels call, echoed by Wing:

Lead: *"Hammer 11, Speed and Angels left"*

Wing: *"Hammer 12, Speed and Angels right"*

Lead will then Check the flight 50° away from Wing:

Lead: *"Check left/right 50"*

Both fighters will select MRT and use G to maintain airspeed to turn 50°. Wing will continue to turn toward Lead and direct a "reverse" when Lead is at Wing's canopy bow.

Wing: *"Reverse"*

Wing will reverse the turn and fly a pure pursuit curve, maintaining altitude, and putting Lead between the airspeed and altitude boxes in the HUD. The throttle will remain at MRT for compressor stall mitigation in the T-45. Wing will then call:

Wing: *"Hammer 12 in from X.X (A/A TACAN DME)"*

Wing will count down the range to Lead in tenths of miles, while directing Lead to either tighten or ease the turn to maintain 40° AOT. At 1.0 DME, Wing will call:

Wing: *"Fox-2"*

Lead will then attempt to defeat the Sidewinder shot by calling:

Lead: *"Chaff Flare"*

At the Chaff Flare call, the Defender will execute a defensive break turn, placing the lift vector on to slightly below the Attacker bleeding down to 300 KIAS. The Defender will then stop fighting and set power to 94–96%. At the Fox-2 call, the Attacker will drive to the Defender's point of departure. A good technique is to use a geo-reference point to drive to the Defender's departure point (e.g. a cloud, ground ref, etc.). When aspect stops changing and the line of sight (LOS) rate rapidly increases, the Attacker is inside the Bubble, and has arrived at the Attack Window.

Upon Bubble entry, the Attacker will execute Attack Window Entry Mechanics. Roll to place the lift vector on to slightly below the Defender, and max perform to stabilize the Defender in your canopy (about 1 fist above the canopy bow). Analyze the Attack Window Entry (AWE) timing and subsequent defender maneuvering. The Attacker will then use lead, lag, and pure pursuit to arrive in the Control Zone while managing range, angles and closure. The Attacker should look to initially take a tracking gunshot (1000 – 1500 ft) on a non-maneuvering, cooperative target. After several successful tracking gunshots, the Defender should be cleared to maneuver by the Attacker's trunk IP calling "cleared to maneuver." The Defender will leave the throttle set and use no more than 17 units AOA while maneuvering to the deck. The Attacker should look to employ tracking gun shots by squeezing the trigger with the pipper 1 – 2 plane lengths in front of Lead, walking the pipper through the spine of the target aircraft, and then lagging to the Control Zone.

The Defender should not be trying to create overshoots or closure problems. A wing flash into the Attacker may be used to indicate poorly managed range, angle, or closure problems to simulate where a reversal would normally have neutralized the fight, but the Defender will not maneuver to create BFM merge mechanics. A KIO call may be initiated by any member of the flight when any aircraft is Bingo fuel, but is normally initiated by the Attacker's trunk IP when training objectives are met, per the TACSOP. For amplifying information regarding follow-on BFM training refer to the BFM FTI.

605. GETTING TO THE CONTROL ZONE AND ATTACK WINDOW ENTRY

After the shot, roll wings-level or ease to the Defender's point of departure and watch the Defender go across the horizon for approximately 2 to 3 seconds. About this time the Attacker should be near the point where the Defender began the defensive break turn (Attack Window Entry). With the Defender at approximately the 11 or 1 o'clock position (roughly the canopy bow), execute Attack Window Entry Mechanics to arrive in the Defender's Control Zone with range, angles, and closure under control. Getting to the Control Zone with angles and closure under control comes from executing a max performance break turn toward the right piece of sky at just the right time.

606. MISSING THE CONTROL ZONE

There are a few reasons why the Attacker may ultimately arrive outside the Defender's Control Zone after executing Attack Window Entry Mechanics. If the Attacker does not max perform the jet, the Defender will initially establish a greater turn rate, such that the Attacker will have to work longer and harder to make up the angles to achieve Control Zone management.

If the break turn was late (after passing beyond the Attack Window), the flight path created may result in passing aft of or outside of the Defender's Control Zone (or both). If early (before the Attack Window), the chance of turning in front of the Post, resulting in an in-close overshoot with the Defender and high AOT, may exist. This subsequent in-close overshoot often gives the Defender an opportunity to reverse and possibly neutralize the fight. If a reversal opportunity is recognized by Lead it will be noted for the debrief, but a high-energy reversal will not be executed.

607. MISALIGNED TURN CIRCLES (MATC)

When Attack Window discussion, the notion of 'near the point where the Defender began the break turn' was used to mean just that – 'near' where the Defender broke, not 'exactly' where the Defender broke. Assuming similarly performing aircraft, if the Attacker executes Attack Window Entry Mechanics at exactly the point in space as where the Defender executed, theoretically the Attacker will end up on the same turn circle as the Defender, but with the nose stuck in lag. Breaking 'near' where the Defender broke (or in the Attack Window), will establish what's needed for 'misaligned turn circles.'

Misaligning turn circles ever so slightly with respect to the Defender's is essential to being able to get the Attacker's nose on the Defender, given there aren't too many angles to make up.

When turn circles are misaligned appropriately, the Attacker will initially notice the Defender drift aft on the canopy while going slightly outside the Defender's turn circle and further in to lag. However, as the fight continues, the Attacker will eventually observe the Defender's forward movement on the canopy and subsequent crossing back inside the Defender's turn circle, with the nose inching towards pure pursuit. When misaligned properly, the Attacker will achieve the desired effect while remaining inside the Control Zone the entire time. Excessively misaligning turn circles will lead to flying too far outside the Defender's turn circle and eventually, out of the Control Zone. Excessively misaligned turn circles are most commonly attributed to late Attack Window Entries.

608. EXCESSIVE RANGE OR STUCK IN LAG

After poorly performed Attack Window Entry Mechanics, the Attacker may become stuck in lag and/or stuck outside of gun range. Executing a Low Yo-Yo, as previously described, will be the solution to overcome excessive range or lag problems. With the Low Yo-Yo, if the Defender has not maneuvered all the way down to the deck, then the Attacker may be able to trade altitude for airspeed, enabling more G (and turn rate) available to make up the remaining angles to close on the Defender. If getting stuck in lag was the result of aligned turn circles, then a Low Yo-Yo will tend to force some misalignment and allow for future shot opportunities.

609. EXCESSIVE CLOSURE OR TOO LITTLE RANGE

If the Attacker is able to get the nose on the Defender and take a shot, some amount of closure on the Defender may have been generated. A little closure can be manageable if the Attacker is still within the Control Zone; too much closure can be a problem if it results in the Attacker flushing out in front of the Control Zone. Some form of a High Yo-Yo or modulation of the throttle should at least allow the Fighter to slow closure, or if aggressive enough, even allow for some range opening.

610. METHODS FOR TAKING GUNSHOTS

Taking the shot is not difficult in theory. The Attacker must put the pipper on the Defender and pull the trigger. Sounds easy enough, but pilots will find that getting the pipper to settle down on the target takes fine control inputs. Below are a couple methods for getting bullets to impact the target:

- ***Saddling In***. By adjusting power, the Attacker may be able to settle comfortably 'in the saddle' (1000-1500 ft) with the pipper stationary on the target. It can be difficult to get the pipper on the target, and equally difficult to hold it there. A disadvantage to this method may occur if the Attacker excessively decreases power to maintain ranging during pure pursuit and ends up at an energy deficit with respect to the Bandit.

- ***Pilot Controlled Error Technique***. In essence the pilot-controlled error technique is accomplished by placing the pipper slightly in front of the target (e.g., a half pipper's width), and pulling the trigger while letting the pipper drift back through the target.

The spread of bullets from this technique forces the target to fly through a string of bullets, increasing the probability of a successful engagement. If the Defender survives, the Attacker should be able to execute a Low Yo-Yo, placing the nose back out in front of the Defender for another quick shot. This method has a couple of advantages: first, pipper control is easier since the Attacker is not trying to steadily hold the pipper on a target. Second, allowing the pipper to drift aft to the target briefly puts the Attacker's nose into a slight lag, allowing for a higher energy state throughout the maneuver.

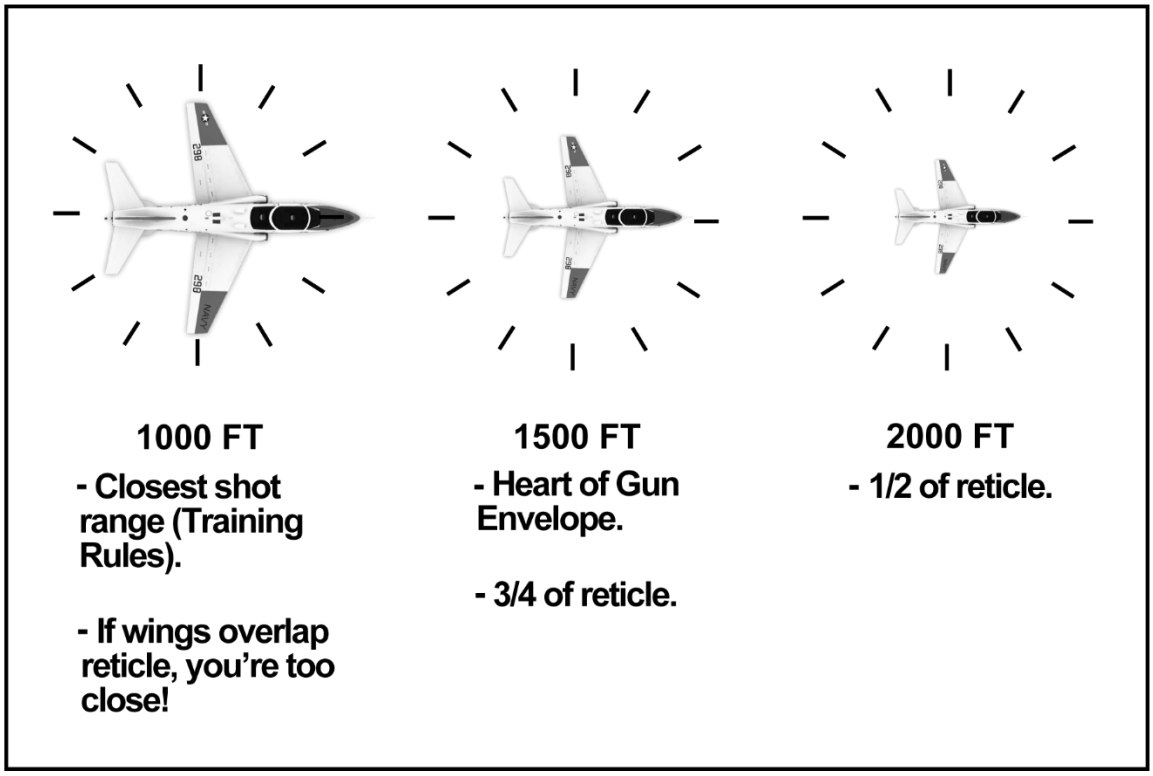

Figure 6-2 The Gun Reticle and Associated Ranges

Common Errors:

1. Poor timing on the Attacker's Attack Window Entry.

2. Inadequate/floated Attack Window Entry Mechanics.

3. Leaving the pipper on the target for too long and generating excessive closure, leading to an in-close overshoot.

4. Bleeding past turn rate numbers accidentally, and not allowing MATC geometry to work.

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CHAPTER SEVEN
DIVISION TACTICAL FORMATION

701. ADMIN

One of the objectives of this part-task training mission is to expose you to a division TACAN or running rendezvous by sections. You may or may not have seen this in your division formation training, but it is a common practice when joining a division if weather precludes a division interval takeoff. You will marshal as a division and check-in IAW SOP/TAC SOP. Taxi and takeoff will be in sections as briefed by the Flight Leads. The Division Lead will brief a rendezvous game plan. This can be accomplished via a running rendezvous or TACAN rendezvous. As -2, fly the briefed formation. As -4, fly the briefed formation off of -3, but be sure to put -3 between your aircraft and the lead section when executing the join.

702. FORMATION

Flying safely and effectively in Division Tactical Formation requires good position keeping off of your Section Lead but also an awareness to the other members of the division. In this stage of training, you will execute Division Wall and Fluid Four. In your follow-on training at the FRS and beyond, you will be introduced to other tactical formations.

703. DIVISION WALL

Division Wall is a flight of four fighters line abreast in combat spread. This formation allows the division to execute simultaneous attacks against air threats or surface targets while maintaining maximum visual mutual support. The center of the formation is -1 and -3 at 0.8-1.0 nm abeam. The outside is -2 and -4 maintaining the same abeam distance from their respective leads. An altitude de-confliction game plan will be briefed by the Division Lead. See Figure 7-1 below for an example altitude de-confliction game plan.

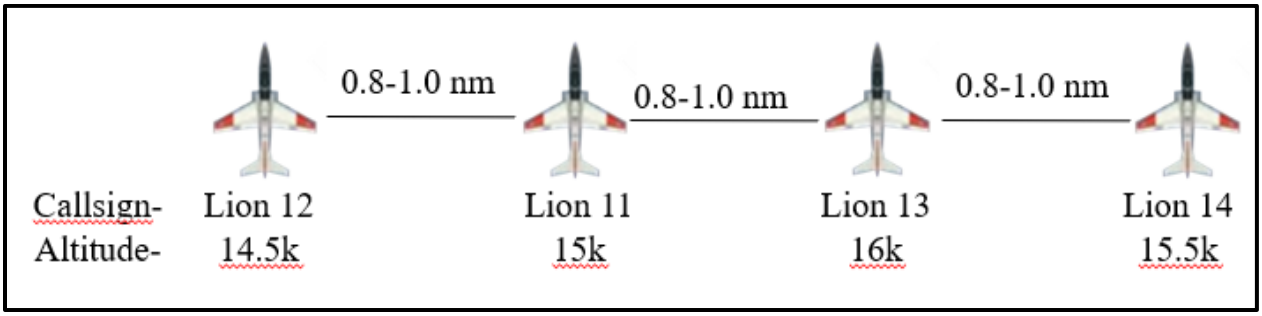

Figure 7-1 Division Wall Position

704. FLUID FOUR

The Fluid Four formation consists of -1 and -3 in combat spread, with their respective wingmen in Tac Wing. This formation is commonly used during the administrative phase of flight to maintain visual mutual support between -1 and -3, but to also allow increased mission

cross-check time and comfort for the wingmen during long transits. If operating in the National Airspace System, a slightly condensed version of this formation can be used to establish the FAA recognized "Standard Formation" or "ATC Spread," in which all aircraft are within 1nm and 100' feet of altitude from Lead. During the mission phase, this formation can be used to hold at a briefed location or specified race-track pattern, quickly change direction, and set a briefed tactical formation. Fluid Four provides an intermediate step between a purely administrative division Fingertip formation and a tactical formation such as Division Wall. An altitude de-confliction game plan will be briefed by the Division Lead, for example, Lion 11 at 16k ft, and Lion 13 at 15k ft. See Figure 7-2 for an example altitude de-confliction game plan.

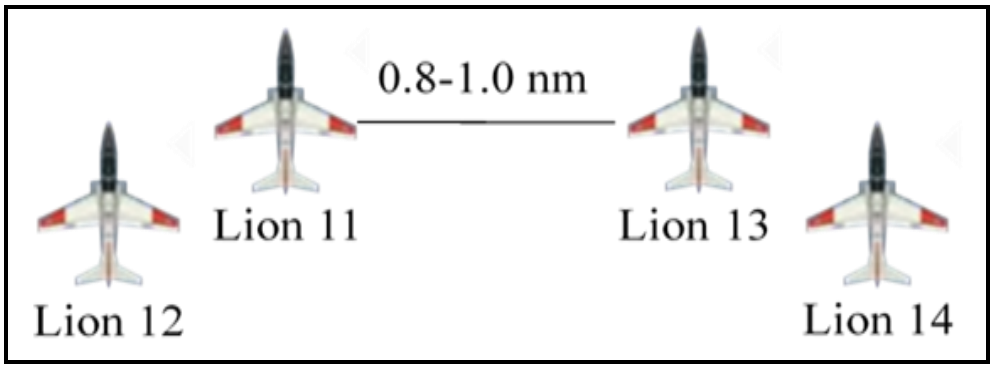

Figure 7-2 Fluid Four Position

705. TAC ADMIN

Division TAC ADMIN in CNATRA is accomplished in much the same manner as a section with a couple exceptions. Once the division is joined, the Division Lead will call a flow heading and deploy the division to a co-altitude Division Wall in preparation for the G-Warm.

Division Lead: *"Lion, flow 360, deploy."*

No response is required from the wingmen before initiating any maneuver. If a transmission is unclear, query the Division Lead for clarification. Of note, as -4, you will need to take a larger cut away and potentially descend to build airspeed to rapidly get to position and not become sucked. You should strive to remain abeam -3 as you push out and then trade airspeed for altitude as you reach the appropriate abeam distance to arrive co-altitude with the division. Additionally, the Master Armament switch will remain in the SAFE position during this stage of training to promote habit patterns consistent with those of your future fleet aircraft.

706. DIVISION G-WARM (WALL FORMATION)

The Division G-Warm will be executed as in section but with four aircraft line abreast. After the G-warm, all flight members will establish their briefed PADS, and the Division Lead will initiate the "fenced in" call. Consult your local TAC SOP for specific guidance.

707. YARDSTICK GAME PLAN

The Division Lead will brief a yardstick game plan. Options include having the fighter's range only within each section, ranging only between -1 and -3, and ranging within sections but with alternate ranging between -1 and -3 quickly available. For example, if you assign -1 40X, -2, 103X, -3 104X, and -4 41X, you will have ranging within the sections but also the ability for -3 to select one digit to the left and range to -1.

708. DIVISION COLLAPSE TO PARADE

During this stage of training, you will practice collapsing the flight from Division Wall to parade. A common occurrence of this procedure in the Fleet is collapsing the flight to a fingertip parade, "loose cruise," or Tac Wing after the G-Warm in order to set up for a spacer pass, or to simply administratively rejoin the flight. Because we are not "Fencing Out," a "cleared to join" call will be used by the Division Lead to initiate the join. ***It is imperative that you execute a professional join.*** You must put your aircraft in a position to execute a join you have been taught, either CV or running rendezvous while maintaining SA to ALL other members of the division. An underrun may or may not be available depending on the position of the other joining members. As -4, you must keep -3 between yourself and -1.

709. FENCE-OUT

Once flight conduct is complete, the Division Lead will direct the flight to fence out. Since division battle damage checks are most commonly executed in section, a simple "fence out" call will instruct the wingmen to join on their respective leads. If the Division Lead desires to join the division, the following call will be made:

Division Lead: *"Lion, fence-out, you are cleared to join."*

710. DIVISION BATTLE DAMAGE CHECKS

When battle damage checks are to be done within a division, they should normally be done in separate sections from a fluid four formation. Another acceptable option is to have -4 check the rest of the division and have -2 check -4.

When conducted as a three-ship, the standard is to execute the checks from fingertip formation with -3 checking -1 and -2, then -2 checking -3.

711. DIVISION TACTICAL REJOIN

If battle damage checks were completed in section, -3 and -4 will join on -1. As -4, simply fly Tac Wing on -3 while -3 joins on -1. Once -3 has stabilized in the briefed formation (parade, admin cruise, or Tac Wing), set the briefed formation off of -3. The Division Lead will initiate the "fenced-out" comm once the join is complete.

712. DIVISION TAC FORM MANEUVERS

Tac turns and in-place turns will be executed from Division Wall and Fluid Four. These turns will be initiated by the Division Lead with no response required from the wingmen. It is imperative that flight members maintain good positioning keeping off of their respective flight lead and execute sound timing with consistent basic air work. MRT G to maintain 330 KIAS will be used for all maneuvers. Of note, when executing maneuvers in Division Wall, strive to be on your assigned altitude. You are cleared to climb or descend to adjust your position once the maneuver is complete.

713. TAC TURNS IN DIVISION WALL

A tac turn in Division Wall is executed in much the same manner as in section. The wingman on the outside of the called turn will simply turn for 90° using MRT, G to maintain 330 KIAS. The next aircraft on the inside of the turn will initiate the turn when the preceding aircraft's intakes are approaching. Each subsequent aircraft will turn when the preceding aircraft's intakes are approaching. Only tac turns for 90° of heading change will be called. Figure 7-3 illustrates the maneuver.

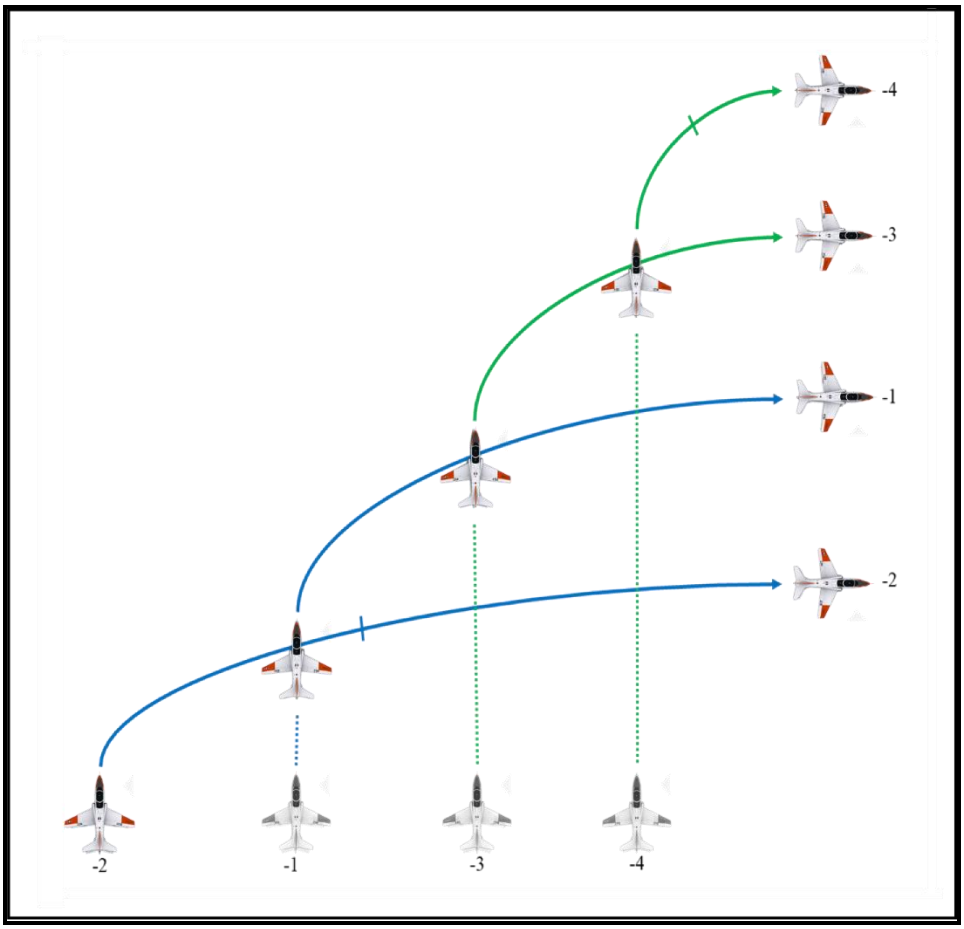

Figure 7-3 Division Wall Tac Turn

714. IN-PLACE TURNS IN DIVISION WALL

In-place turns in Division Wall are generally used to egress a target area after executing simultaneous air-to-surface target attacks. Execution is identical to that of a section in-place turn but with four aircraft line abreast. Only turns requiring 180° of heading change will be called.

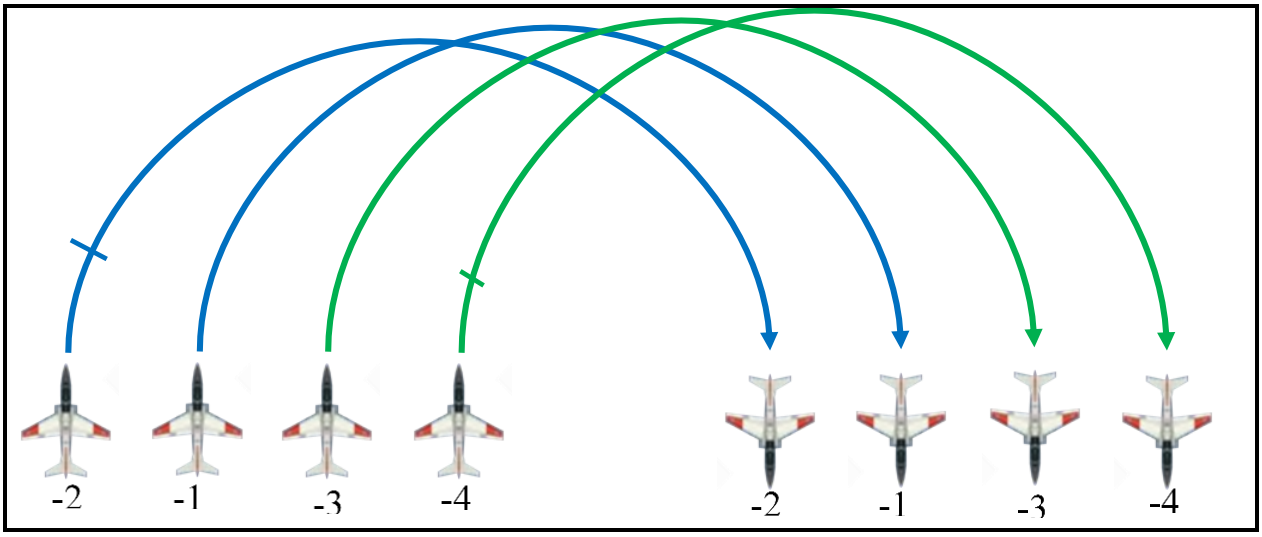

Figure 7-4 Division Wall In-Place Turn

715. TAC TURNS IN FLUID FOUR

A tac turn in fluid four simply maneuvers the division through 90° of heading change. Depending on the direction called and formation, either -1 or -3 will initiate a MRT, G to maintain 330 KIAS turn into the other Section Lead while noting the position of the wingman on that side of the formation to ensure safe flight path de-confliction. The wingman in the first section to maneuver will simply fly Tac Wing off their Section Lead and flow to the inside of the turn. The second Section Lead to maneuver will execute a standard tac turn when the first Section Lead's intakes are approaching. The second wingman will fly tac wing off their Section Lead utilizing energy and geometry to maintain Tac Wing position throughout the turn but end up on the outside of the formation looking through their Section Lead to gain sight of the other section. Figure 7-5 illustrates the movements of the wingmen. As wingmen, it is your goal to maintain position off of your Section Lead but also maintain sight of the other section to the maximum extent practical. Only tac turns for 90° of heading change will be called.

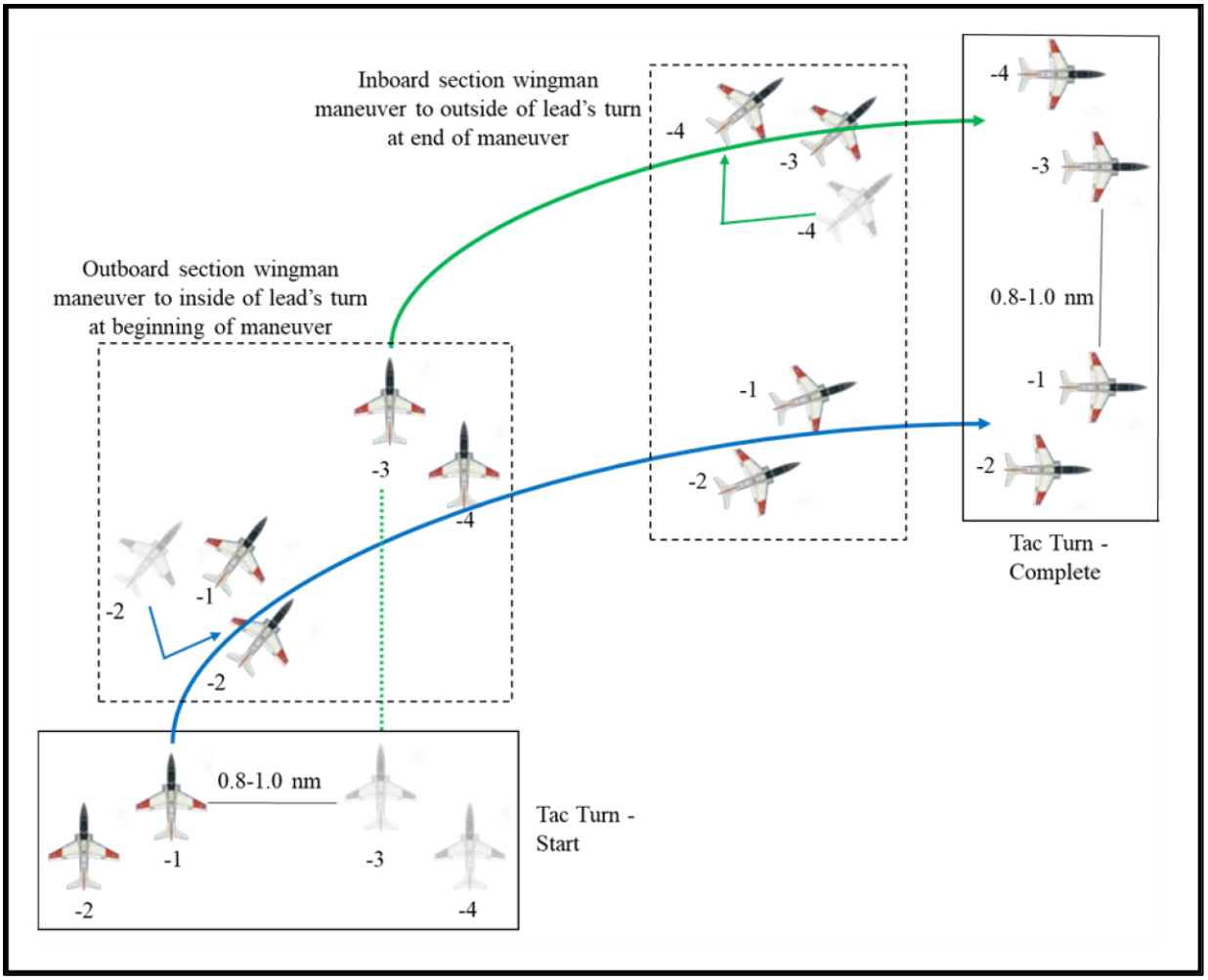

Figure 7-5 Fluid Four Tac Turn

716. IN-PLACE TURNS IN FLUID FOUR

An in-place turn in fluid four simply maneuvers the division through 180° of heading change. Dash 1 and -3 simply execute an in-place turn. The wingman on the outside of the turn will maneuver to the inside of the turn in Tac Wing. The wingman on the inside of the turn will maintain Tac Wing position but will need to manage energy and geometry to roll out on the outside of the formation looking through the Section Lead to regain sight of the other section. Figure 7-6 illustrates initial formation and desired outcome with arrows to show the movements of the wingmen. Only in-place turns for 180° of heading change will be called.

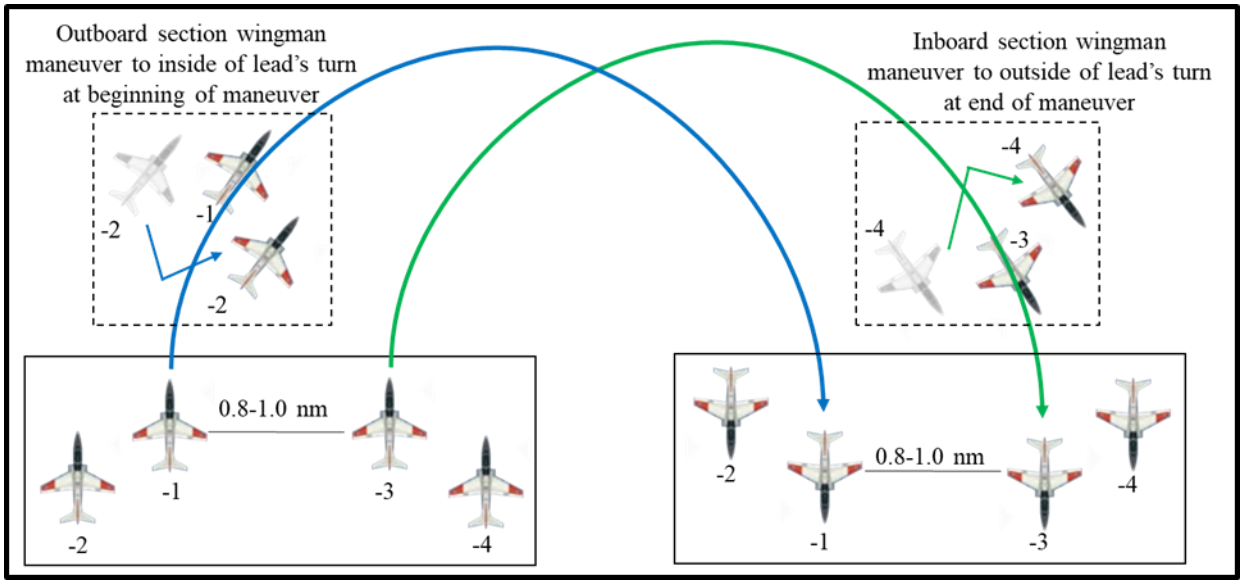

Figure 7-6 Fluid Four In-Place Turn

717. TYPICAL STUDENT FLIGHT SEQUENCE

Conduct at discretion of Flight Lead
Division deploy to wall formation followed by Division G-warm
FENCE-in/set PADS (wall formation with altitude de-confliction)
    Tac turns and In-place turns
Collapse to Parade / Division battle damage checks
Deploy to wall formation (desired)
    Tac turns and In-place turns
Collapse to Admin Cruise (desired)
Deploy to fluid four formation
    Tac turns and In-place turns
Battle damage checks (section)
Division tactical rejoin

718. TYPICAL IUT FLIGHT SEQUENCE

Conduct at discretion of Flight Lead
Division deploy to wall formation followed by Division G-warm
FENCE-in/set PADS (wall formation with altitude de-confliction)
    Tac turns and In-place turns
Collapse to Parade
Deploy to fluid four formation
    Tac turns and In-place turns
Division lead change in fluid four (swap -1 and -2 with -3 and -4)
TACAN rendezvous by section
Repeat conduct and recover

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